



Estd 2008

BABA INSTITUTE OF TECHNOLOGY AND SCIENCES

(Approved by AICTE, New Delhi and Affiliated to JNTU, Kakinada)

Formerly known as VISAKA ENGINEERING COLLEGE

Accredited by NAAC

Bakkannapalem (V), Madhurawada (PO), Visakhapatnam-530048.

3.3.2 Number of research papers in the journals notified on UGC website during the last five years (10)

S.NO	ACADEMIC YEAR	No. of research papers in the journals notified on UGC website during the last five years
1	2020-2021	20
2	2019-2020	25
3	2018-2019	43
4	2017-2018	59
5	2016-2017	32
	Total	179

Principal

PRINCIPAL
Baba Institute of Technology and Sciences
Bakkannapalem (V), Madhurawada (P)
Visakhapatnam

☎ Tele - Fax : 0891-2568811, Off : 0891- 2569933, Cell : 8008844699

Website : www.bitsvizag.com, e-mail: principal@bitsvizag.com

Under the Management of **AMBEDKAR EDUCATIONAL & WELFARE SOCIETY** Regd : 174 of 2005

ANALYSIS OF PROCESS CAPABILITY ABOUT TURNING CENTER

Chundru.Venkata Rao¹

N.Tejeswar Reddy¹

B.Sai Prasanth¹

Pujari Srinivasa Rao²

P.Thirumala Rao¹

Abstract

Analysis of process capability convinces that processes need to be according to manufacturing industry specification, while reducing the process variation and improving the product quality characteristic. The aim is to verify the performance of the process and also the ability of the machine to achieve within the tolerance limit specified. 100 sample shafts were turned on the turning center with subgroup size being ten. The performance needs to be within the control limits and also capable of meeting up to specification. Further, the capability index (C_p) measured in this case is greater than 1. Also, the machine capability (C_{pk}) for this manufacturing application is also greater than 1, thus making the process capable.

Keywords:

Turning center;
Control charts;
Process capability;
Machine ability.

Copyright © 2017 International Journals of Multidisciplinary Research Academy. All rights reserved.

Author correspondence:

Chundru.Venkata Rao

Department of Mechanical Engineering,
Baba Institute of Technology and sciences,
Visakhapatnam.

1. Introduction

Nowadays, most of the manufacturing companies spend a lot of money for machine tools with advanced technology such as CNC tools and automated machine tools. These will help to improve the quality and dispatch time of the products that they produced. The investors require that the machines they purchase have high flexibility, good running condition and highly precision-oriented turning centers. In the manufacturing sector, the accuracy of parts is very important because the accuracy of the finish part helps in evaluating the machine's accuracy that produces the particular part. In the early 1970s the concept of process capability was introduced by Juran, an expert in world quality. Subsequently, for almost 40 years many accepted indices such as C_p , C_{pk} and C_{pm} were developed to measure the performance of the respective process [1]. The measure of the uniformity of performance is process capability. Variability of the process output is measured by its uniformity in process product. The relative deviation in the product specification and requirements is nothing but variability in process product. Process capability analysis is a technique applied in many stages of the product cycle. This includes product design, manufacturing planning process and production in machines for each operation. This technique helps in determining the ability for achieving the engineering values and manufacturing parts within the tolerance limits.

¹ Department of Mechanical Engineering, Baba Institute of Technology and Sciences, Visakhapatnam

² Department of Mechanical Engineering, GITAM – Visakhapatnam

The long term performance level of a process which is brought under statistical control is called process capability. Statistical process control is a quality assurance tool which is used to improve the quality of manufacture and helps in scoring the end-customer satisfaction [2]. The ability of the combination of the

Model Analysis of Gas Turbine Blade with Different Metals by Using Ansys 11.0

P.V.Durga Prasad¹
S.Polinaidu^{2*}
R.Haranath¹
Ch.Venkata Rao¹

Abstract

The gas turbine obtains its power through a hot fluid with high temperature and pressure moving at a high speed and expanding through the several rings of moving and fixed blades. The purpose of gas turbine is mainly to generate electricity and to propel different machinery and objects via the mechanical energy produced. Gas turbines use up high pressure burnt gas to produce electricity and industrial application and is also used to propel jet engines. The number of gas turbine stages have a more effect on the turbine blades and are designed for each stage. Most of the gas turbine engines are double spool designs, meaning that there is a low pressure spool and a high pressure spool and the turbines used three spools, intermediate pressure spool is added in between the low and high pressure spool.

Keywords:

Turbine blades,

Graphite,

Titanium,

Ansys

The increase of turbine rotor inlet temperature, thermal efficiency and power output of gas turbine increases and rotor inlet temperature of gas turbine is far above the melting point of the blade material. CATIA is used for design of solid model for the analyses of gas turbine blade, further analysis of this solid model in ANSYS software by generating finite element model, and then apply boundary condition; this includes the specific post-processing and life assessment of turbine blade. The program makes effective utilize of the ANSYS pre-processor to mesh complicated turbine blade geometries and apply boundary conditions. Designing of a turbine blade is in CATIA with the assist of co-ordinate generated and to demonstrate the pre-processing capabilities, static and dynamic stress analysis results and life assessment on CMM. In these analyses to study the thermal stresses on turbine blade with titanium and graphite as material and obtain the natural frequencies of the turbine blade. A structural analysis has been carried out to study the stresses, shear stress and displacements of the turbine blade which is been developed due to the combination effect of centrifugal and thermal loads. An attempt is also made to suggest best material through comparing the results obtained from the analyses of two different materials (Graphite and Titanium) of a turbine blade. Titanium is the best material based on the analysis results and plots, as well as it has high-quality material properties at higher temperature as compare to that of graphite material.

Author correspondence:

S.Polinaidu
Department of Mechanical Engineering,
Thandra Paparaya Institute of science & Technology,
Bobbili,
vizayanagaram, India ; Email:

¹Department of Mechanical Engineering, MVSR Engineering College, Nadargul, Hyderabad, India.

²Department of Mechanical Engineering, Thandra Paparaya Institute of science & Technology, Bobbili, vizayanagaram.

³Department of Mechanical Engineering, Baba Institute of Technology and Sciences, Visakhapatnam, India.

1. INTRODUCTION

PRINCIPAL
Baba Institute of Technology and Sciences
Bakkannapalem (V), Madhurawada (P)
Visakhapatnam

Chapter 39

Fabrication and Testing of Natural Fibre-Reinforced Hybrid Composite Plate with Hemp and Pineapple



Dharmala Venkata Padmaja, Ch. Uma Chaitanya, Tankala Ravi Teja and S. Srikan

Abstract This paper presents an experimental report on the fabrication of natural fibre composite material. Natural fibres Hemp and Pineapple were used as reinforcement and epoxy resin and hardener mixed in 10:1 ratio, respectively, was used as matrix. Mechanical properties like tensile, breaking load, elongation, and impact strengths were studied for composites made of these natural fibres. Retting and manual process were used to extract the natural fibres. Hybrid composites were prepared using Hemp/Pineapple fibres of 0/40, 15/25, 20/20, 25/15, and 40/0 Weight fraction ratios, while overall weight of the epoxy resin was considered as 60% of total weight, and fibre weight fraction was fixed as 40% of total weight. The fabricated specimens were made and tested as per ASTM standards for finding the mechanical properties.

Keywords Natural composite fibre · Hemp and Pineapple · Hybrid composites and mechanical properties

39.1 Introduction

Fibre type is commonly classified based on its origin: plant, animal, or mineral. All animal fibres contain protein as their major structural component, whereas plant fibres mainly consist of cellulose. In spite of the fact that mineral-based natural fibres present within the asbestos group of minerals and were once utilized extensively in composites, these are now eluded due to associated health problems (carcinogenic through inhalation/ingestion) and are prohibited in many countries

D. V. Padmaja (✉) · Ch.U. Chaitanya · T. R. Teja
Department of Mechanical Engineering, Baba Institute of Technology and Sciences,
Visakhapatnam, India
e-mail: padmajadv@gmail.com

S. Srikan
Department of Mechanical Engineering, Lendi Institute of Engineering and Technology,
Visakhapatnam, India

© Springer Nature Singapore Pte Ltd. 2019
S. Pujari et al. (eds.), *Recent Advances in Material Sciences*,
Lecture Notes on Multidisciplinary Industrial Engineering,
https://doi.org/10.1007/978-981-13-7643-6_39


PRINCIPAL
Baba Institute of Technology and Sciences
Bakkannapalem (V), Madhurawada (P)
Visakhapatnam

Chapter 39

Fabrication and Testing of Natural Fibre-Reinforced Hybrid Composite Plate with Hemp and Pineapple



Dharmala Venkata Padmaja, Ch. Uma Chaitanya, Tankala Ravi Teja
and S. Srikan

Abstract This paper presents an experimental report on the fabrication of natural fibre composite material. Natural fibres Hemp and Pineapple were used as reinforcement and epoxy resin and hardener mixed in 10:1 ratio, respectively, was used as matrix. Mechanical properties like tensile, breaking load, elongation, and impact strengths were studied for composites made of these natural fibres. Retting and manual process were used to extract the natural fibres. Hybrid composites were prepared using Hemp/Pineapple fibres of 0/40, 15/25, 20/20, 25/15, and 40/0 Weight fraction ratios, while overall weight of the epoxy resin was considered as 60% of total weight, and fibre weight fraction was fixed as 40% of total weight. The fabricated specimens were made and tested as per ASTM standards for finding the mechanical properties.

Keywords Natural composite fibre · Hemp and Pineapple · Hybrid composites and mechanical properties

39.1 Introduction

Fibre type is commonly classified based on its origin: plant, animal, or mineral. All animal fibres contain protein as their major structural component, whereas plant fibres mainly consist of cellulose. In spite of the fact that mineral-based natural fibres present within the asbestos group of minerals and were once utilized extensively in composites, these are now eluded due to associated health problems (carcinogenic through inhalation/ingestion) and are prohibited in many countries

D. V. Padmaja (✉) · Ch.U. Chaitanya · T. R. Teja
Department of Mechanical Engineering, Baba Institute of Technology and Sciences,
Visakhapatnam, India
e-mail: padmajadv@gmail.com

S. Srikan
Department of Mechanical Engineering, Lendi Institute of Engineering and Technology,
Visakhapatnam, India

© Springer Nature Singapore Pte Ltd. 2019
S. Pujari et al. (eds.), *Recent Advances in Material Sciences*.
Lecture Notes on Multidisciplinary Industrial Engineering.
https://doi.org/10.1007/978-981-13-7643-6_39

8481
PRINCIPAL
Baba Institute of Technology and Sciences
Bakkannapalem (V), Madhurawada (P)
Visakhapatnam

Chapter 39

Fabrication and Testing of Natural Fibre-Reinforced Hybrid Composite Plate with Hemp and Pineapple



Dharmala Venkata Padmaja, Ch. Uma Chaitanya, Tankala Ravi Teja and S. Srikan

Abstract This paper presents an experimental report on the fabrication of natural fibre composite material. Natural fibres Hemp and Pineapple were used as reinforcement and epoxy resin and hardener mixed in 10:1 ratio, respectively, was used as matrix. Mechanical properties like tensile, breaking load, elongation, and impact strengths were studied for composites made of these natural fibres. Retting and manual process were used to extract the natural fibres. Hybrid composites were prepared using Hemp/Pineapple fibres of 0/40, 15/25, 20/20, 25/15, and 40/0 Weight fraction ratios, while overall weight of the epoxy resin was considered as 60% of total weight, and fibre weight fraction was fixed as 40% of total weight. The fabricated specimens were made and tested as per ASTM standards for finding the mechanical properties.

Keywords Natural composite fibre · Hemp and Pineapple · Hybrid composites and mechanical properties

39.1 Introduction

Fibre type is commonly classified based on its origin: plant, animal, or mineral. All animal fibres contain protein as their major structural component, whereas plant fibres mainly consist of cellulose. In spite of the fact that mineral-based natural fibres present within the asbestos group of minerals and were once utilized extensively in composites, these are now eluded due to associated health problems (carcinogenic through inhalation/ingestion) and are prohibited in many countries

D. V. Padmaja (✉) · Ch.U. Chaitanya · T. R. Teja
Department of Mechanical Engineering, Baba Institute of Technology and Sciences,
Visakhapatnam, India
e-mail: padmajadv@gmail.com

S. Srikan
Department of Mechanical Engineering, Lendi Institute of Engineering and Technology,
Visakhapatnam, India

© Springer Nature Singapore Pte Ltd. 2019
S. Pujari et al. (eds.), *Recent Advances in Material Sciences*,
Lecture Notes on Multidisciplinary Industrial Engineering.
https://doi.org/10.1007/978-981-13-7643-6_39

481
PRINCIPAL
Baba Institute of Technology and Sciences
Bakkannapalem (V), Madhurawada (P)
Visakhapatnam

Experimental Data of Calophyllum Inophyllum Bio Diesel Effect by Varying Compression Ratio and Injection Pressure on Emissions

D. V. Padmaja*

M.Sirisha**

T. Satyanarayana***

ABSTRACT

Keywords:

Biodiesel,
Transesterification,
VCR Engine,
Injection pressure,
compression ratio,
Engine emission.

Degrading fossil fuel resources have thrown world into chaos, besides air pollution and green house gases are a major concern. The fossil fuel shortage in near future is inevitable. This situation triggered the awareness to find alternative and sustainable energy resources. Biodiesel is a better solution as its properties are close to petrodiesel. In this paper, Biodiesel is produced from Calophyllum Inophyllum oil by two stage transesterification process using acid and base catalysts. Acid esterification is done by mixing 1%(by wt. of oil) of 99% pure anhydrous sulphuric acid in 1000 ml of oil with oil to methanol ratio 6:1 and maintained at a temperature of 55° C for 120 minutes. This reduces the acid value of the oil. Now the oil is subjected to base catalyst. 1% of KOH (by wt. of oil) is added to a mixture of oil to methanol ratio 4:1. The mixture is stirred for two hours at 60° C. The yield is about 89%.

Due to high viscosity it is difficult for regular engines to inject biodiesel into combustion chamber. This results in improper mixing of fuel and reduces ignition quality. In this paper the fuel injection pressure has been increased and compression ratio has been varied and tested under different operating conditions. The increase in injection pressure resulted in better atomization of fuel and better mixing with the air in combustion chamber. The increase in compression ratio resulted in higher compression pressure which ensures complete combustion of fuel in the chamber. These modifications result in reduction of emissions and smoke up to about 10% compared to normal engines running on biodiesel.

Author correspondence:

D. V. Padmaja

Assistant Professor, Department of Mechanical Engineering

Blts, Vizag, India.

1. Introduction:

DESIGN AND FABRICATION OF INDUSTRIAL PARAMETER MONITORING AND CONTROL ROBOT

CH Uma Chalthanya*

K Sri Ram Murthy**

P Silva Vasu***

D Rama Krushna****

K Yuva Kishore*****

K P Bala Krishna*****

Abstract

Now-a-days the accidents in the coal mine industries have increased. Even if any explosion occurs it can't be easily known to the worker and it may cause accidents. So in order to avoid this, a robot has been designed and this robot is allowed to monitor the ambient situations inside the coal mine industry. Some of the environmental parameters such as methane leakage, temperature, oxygen are sensed by using the high end sensors and the sensed data are transmitted for monitoring the status of the coal mine and to control the robot movement.

Keywords:

DC Motors;
Sensors;
Buzzer;
Battery;

If the temperature exceeds a threshold, the cooling fan is automatically set to ON and if any gas leakage is detected the workers are given alert through a buzzer. By this the human intervention can be avoided inside the industry and the accidents can be prevented.

Author correspondence:

CH Uma Chalthanya, Assistant Professor, Department Of Mechanical Engg, BITS-Vizag, India
Email: uma.nltw@gmail.com

* Doctorate Program, Linguistics Program Studies, Udayana University Denpasar, Bali-Indonesia (9 pt)

** STIMIK STIKOM-Bali, Renon, Denpasar, Bali-Indonesia

*** English Language Specialist, Oller Center, Carriage House, 2nd Floor, California, USA

**** Head, Dept. of Political Science, Faculty of Arts & Culture, South Eastern University, Sri Lanka

***** Head, Dept. of Political Science, Faculty of Arts & Culture, South Eastern University, Sri Lanka

PRINCIPAL
BITS Institute of Technology and Sciences
Bakkannapalem (V), Madhurawada (P)
Vizakhapatnam

Effect on the Performance of the Nano-Particulate Graphite Lubricant in the Turning of AISI 1040 Steel under Variable Machining Conditions

S. Srikanth, Dharmala Venkata Padmaja, P. N. L. Pavani, R. Pola Rao, K. Ramji

Abstract—Technological advancements in the development of cutting tools and coolant/lubricant chemistry have enhanced the machining capabilities of hard materials under higher machining conditions. Generation of high temperatures at the cutting zone during machining is one of the most important and pertinent problems which adversely affect the tool life and surface finish of the machined components. Generally, cutting fluids and solid lubricants are used to overcome the problem of heat generation, which is not effectively addressing the problems. With technological advancements in the field of tribology, nano-level particulate solid lubricants are being used nowadays in machining operations, especially in the areas of turning and grinding. The present investigation analyses the effect of using nano-particulate graphite powder as lubricant in the turning of AISI 1040 steel under variable machining conditions and to study its effect on cutting forces, tool temperature and surface roughness of the machined component. Experiments revealed that the increase in cutting forces and tool temperature resulting in the decrease of surface quality with the decrease in the size of nano-particulate graphite powder as lubricant.

Keywords—Solid lubricant, graphite, minimum quantity lubrication, nanoparticles.

I. INTRODUCTION

METAL cutting results in the generation of a lot of heat due to the plastic deformation of work material, friction at the tool-chip interface and friction between the clearance face of the tool and work piece, which adversely affects the quality of the component produced [1], and hence, the quality of the work piece deteriorating [2].

The use of cutting fluids for heat dissipation [3] is a common practice but is not advisable in terms of cost, and environmental concerns, etc. Its use cannot be entirely eliminated but can be substituted by finding out alternate means and by using bio-degradable fluids [4]. Some other alternatives are by using cryogenic coolants [5], [6] or machining using the concept of Minimum Quantity

Lubrication (MQL) by the application of powders mixed in water and other carrying mediums [7], [8] or by using environmentally friendly lubricants made from vegetable oils extracted from Rapeseed, Corn or Soya bean oil, esters etc., which has been found out to decrease the average temperature at the chip-tool interface [19].

With the advancement in tribology, the use of solid lubricants in machining has been observed during the past few years. It is found that solid lubricants could effectively control the temperature at the machining zone. Experiments also revealed the reduction in surface roughness value due to the use of solid lubricants [21]. Some of the common solid lubricants are graphite, molybdenum disulphide, tungsten disulphide and calcium fluoride which belong to a special class of materials known as lamellar solids [9], [20]. Several studies and researches related to the lubrication properties of Graphite [10], [11] have been carried out over the past few decades and investigated its use as a lubricating medium in various machining operations [22]. Shaji et al. and Venu Gopal et al. have used graphite as lubricant in grinding operations to reduce the heat generated at the grinding zone and found that graphite as lubricant proved to be better than conventional flood coolant in terms of decreased cutting forces, temperature and surface roughness values. It is also found [10] that with the use of graphite as lubricant, the tangential forces decrease and so the reduction in specific energy. Graphite powder, mixed with SAE 40 oil was also used for experimentation purpose as lubricant while machining hardened steel [20], which revealed better machining performance when compared to conventional cutting fluids and dry machining for the selected tool-work combination and cutting condition. One of the limitations of using dry graphite powder in the machining process is that the lubricant powder has to be sprayed continuously into the tool-work piece interface, which does not represent a viable lubrication alternative for industries. To nullify this problem, researchers [12]-[14], [25] are using vegetable oils and petroleum based oils like SAE 40 as a carrying medium to spray the lubricating powders into the tool - work piece interface, which is found to be a viable solution.

Rao et al. [9] have used solid lubricants and analysed the effect and behaviour of various machining parameters with respect to the particle size of the solid lubricant. Graphite particles with 50 micron, 100 micron, 150 micron and 200 micron sizes were used in the form of powder spray during the machining of hardened steel. It was observed from the results

S. Srikanth (Professor) is with the Department of Mechanical Eng., Lendi Inst. Of Eng. & Tech., Vizianagaram, Andhra Pradesh, India (e-mail: ssrikanth@gmail.com).

Dharmala Venkata Padmaja (Asst. Professor) is with the Department of Mechanical Eng., Baba Inst. of Tech. and Sciences, Visakhapatnam, Andhra Pradesh, India (e-mail: padmajadv@gmail.com).

P. N. L. Pavani (Asst. Professor) and R. Pola Rao (Associate Professor) are with the Department of Mechanical Eng., GMR Institute of Technology, Rajamahendravaram, Andhra Pradesh, India (e-mail: pavani.pnl@gmit.org, rpolarao@gmail.com).

K. Ramji (Professor) is with the Department of Mechanical Engineering, A.U. College of Engineering (A), Andhra University, Visakhapatnam, Andhra Pradesh (e-mail: ramjidme@gmail.com).

PRINCIPAL

Baba Institute of Technology and Sciences
Bakkannapalem (M), Madhurawada (P)
Visakhapatnam

Design and fabrication of Three Axes Pneumatic Trailer for shipping Goods

Dharmala Venkata Padmaja*

Yalagada Aravind Kumar**

Kallepalli Rambabu***

Pradhan Hemanta Kumar****

Jami Rajesh Babu*****

Mandangi Dilip Kumar***

Abstract

The objective of this paper is to evaluate the difficulty in unloading the materials. In this consideration, several automobiles garages have concede the facts that, some of the methods, which were most difficult, were endorsed in unloading the materials from the trailer. The loading and unloading of material using trailer is only in one single direction. Moreover, loading and unloading of material, in small compact streets and small roads, is done with high difficulty.

Now our project has mainly concentrated to rectify the loading and unloading of the trailer in all the three sides very easily with suitable arrangement of design. The loading of vehicle from the trailer can be done in three axes by activating the direction control valve, without the application of any impact forces. Compressed air is used in controlling the valve. The trailer cabin is lifted, by using the ram of the pneumatic cylinder as lifter. The compressor engine is coupled with automobile engine drive, so that, when vehicle is running, compressed air is stored. Pneumatic cylinder is activated by using the compressed air, when the valve is activated.

Keywords:

Trailers

Control Valve

Pneumatic Cylinder

Compressed air

Copyright © 2017 International Journals of
Multidisciplinary Research Academy. All rights reserved.

Author correspondence:

Dharmala Venkata Padmaja

Assistant Professor, Department of Mechanical Engineering, Bits, Vizag, India.

* Assistant Professor, Department of Mechanical Engineering, BITS, Vizag, India.

** UG Scholars, Department of Mechanical Engineering, BITS, Vizag, India.

1. Introduction

In building sites, the vehicle used in carry materials is dumper. Dumpers are differentiated from dump trucks by structure: dump truck is having a cab in front of the load, while an open 4-wheeled vehicle with the load skip in front of the driver is a dumper. The name dumper has come from "The skip can tip to dump the load". They are usually powered with diesel. For the secondary use as a site tractor, a towing eye is fitted.

PRINCIPAL

Baba Institute of Technology and Sciences

Bakkannapalem (V), Madhurawada (P)

Visakhapatnam

Experimental Data of Calophyllum Inophyllum Bio Diesel Effect by Varying Compression Ratio and Injection Pressure on Emissions

D. V. Padmaja*

M. Sirisha**

T. Satyanarayana***

ABSTRACT

Keywords:

Biodiesel,
Transesterification,
VCR Engine,
Injection pressure,
compression ratio,
Engine emission.

Degrading fossil fuel resources have thrown world into chaos, besides air pollution and green house gases are a major concern. The fossil fuel shortage in near future is inevitable. This situation triggered the awareness to find alternative and sustainable energy resources. Biodiesel is a better solution as its properties are close to petrodiesel. In this paper, Biodiesel is produced from Calophyllum inophyllum oil by two stage transesterification process using acid and base catalysts. Acid esterification is done by mixing 1% (by wt. of oil) of 99% pure anhydrous sulphuric acid in 1000 ml of oil with oil to methanol ratio 6:1 and maintained at a temperature of 55° C for 120 minutes. This reduces the acid value of the oil. Now the oil is subjected to base catalyst. 1% of KOH (by wt. of oil) is added to a mixture of oil to methanol ratio 4:1. The mixture is stirred for two hours at 60° C. The yield is about 89%.

Due to high viscosity it is difficult for regular engines to inject biodiesel into combustion chamber. This results in improper mixing of fuel and reduces ignition quality. In this paper the fuel Injection pressure has been increased and compression ratio has been varied and tested under different operating conditions. The increase in injection pressure resulted in better atomization of fuel and better mixing with the air in combustion chamber. The increase in compression ratio resulted in higher compression pressure which ensures complete combustion of fuel in the chamber. These modifications results in reduction of emissions and smoke up to about 10% compared to normal engines running on biodiesel.

Author correspondence:

D. V. Padmaja

Assistant Professor, Department of Mechanical Engineering
Bits, Vizag, India.

1. Introduction:

NUMERICAL AND EXPERIMENTAL ANALYSIS OF THERMAL PERFORMANCE IN A PARALLEL FLOW AND COUNTER FLOW DOUBLE PIPE HEAT EXCHANGE

1. Hepsiba Seeli^{*}
2. D. Bhaskar Chandra Rao^{**}
3. K. Lakshmana Rao^{***}
4. Rajesh Ramesh^{****}

Abstract

Heat transfer is taken into consideration as transfer of thermal energy from bodily frame to another. It is the most essential parameter to be measured because the overall performance and efficiency of the double pipe heat exchanger. By the use of CFD simulation software, it reduces the time and operation cost in comparison to experimental calculations. The goal of this paper is to evaluate the influence of the Double pipe heat exchanger to get better heat transfer. The purpose of this examine is to apply CFD software program and Experimental setup to analyze the Temperature drop, Pressure drop and Friction factors, by varying under different inlet conditions like Temperature and Flow rate as a function of both inlet velocity and temperature variations and converting heat exchanger tube material properties like copper and aluminium.

In this experiment, heat transfer from hot fluid to cold fluid with the aid of Double pipe heat exchanger is experimentally by using different organic fluids like Benzene, Glycol, Transformer oil, Acetone and Water to get better heat transfer, the identical thing is established in CFD analysis. The test is accomplished at Laminar flow beneath different flow preparations like Parallel flow and Counterflow.

The test is done on double pipe heat exchanger under various working fluids with different operating conditions, it is predicted that Counterflow exhibits better heat transfer than Parallel flow.

Keywords:

Heat transfer, Double pipe heat exchanger, CFD simulation software, Organic fluids, Copper, Aluminium, Laminar flow, Turbulent flow, Parallel flow and Counterflow

Copyright © 2017 International Journals of Multidisciplinary Research
Academy All rights reserved

Author correspondence:

Hepsiba Seeli,
Assistant Professor, Mechanical Department,
Baba Institute of Technology and Sciences, BITS VIZAG

^{*} Assistant Professor, Mechanical Department, Baba Institute of Technology and Sciences

^{**} Aditya Institute of Technology and Management, Tekkali_532201

^{***} Aditya Institute of Technology and Management, Tekkali_532201

^{****} Research Scholar Andhra University


PRINCIPAL
Baba Institute of Technology and Sciences
Bakkannapalem (V), Madhurawada (P)
Visakhapatnam

Design and Methodology of Line Follower Automated Guided Vehicle for Material Handling

SK. Hyderali*

Y. Siva prasad**

CH. Naveen Kumar***

B. Prasad****

V. Dhanunjaya Rao*****

L. Sai Kumar*****

Abstract

Advancement of automated/computerized guided vehicle assumes a noteworthy part in designing businesses to enhance the material taking care of procedure for late year. In this paper, it is focus on the design and different methodology of line follower automated guided vehicle (AGV) systems. This paper provides an overview on line follower AGV discusses recent technological developments. The essential components of line follower robot and their modification are described in this paper.

Keywords:

Material handling;
Automation;
Line follower agv;
AGV.

Author correspondence:

SK. Hyderali,

ASST. PROFESSOR, Department of Mechanical Engineering, Baba Institute of Technology and Sciences (BITS),
Visakhapatnam (Dist.)

Email: hyderali.mech@gmail.com

1. INTRODUCTION

With the technological advancement in the field of machineries, there have been various attempts to improve the material handling techniques. AGV (Automated Geared Vehicle) is one of the remarkable machine which helps in various tasks such as fork lifting objects, towing, product transportation etc., without the continuous monitoring of human. An AGV works with the simultaneous processing of various parts. The control gadget which is basic to both the driving gadget and exchange gadget works the vehicle and keeps up a definitive

PRINCIPAL

Baba Institute of Technology and Sciences
Bakkannapalem (V), Madhurawada (P)
Visakhapatnam

EVALUATION OF THERMAL INDUSTRIAL AIR HEATERS BASED ON SOLAR SYSTEM

Y.Prasadkumar *

SK.Hyderali**

Abstract

The usage of solar energy heating systems increased greatly during the last decade. Need to make a setup which acquires much of the Sun's radiation as per the output requirement. More than one hundred years ago, black painted water tanks were used as simple solar water heaters in a number of countries. As the development takes place total setup gets advanced. Our main objective is to keep the system in such a way that, yields a better result. The major equipment includes collector, heat-exchanger, storage tank, and pumps & pipes.

Keywords:

Solar energy;
Solar water heaters;
Collector;
Heat-exchanger;
Storage tank.

Author correspondence:

First Author,
Y.Prasadkumar

M.Tech Student Department of *Mechanical Engineering*, Baba Institute of Technology and Sciences (BITS), Visakhapatnam (Dist.)

Second Author,
SK.Hyderali

M.Tech Student Department of *Mechanical Engineering*, Baba Institute of Technology and Sciences (BITS), Visakhapatnam (Dist.)

PRINCIPAL

Baba Institute of Technology and Sciences
Bali-Indonesia (P), Madhurawada (P)
Visakhapatnam

* Doctorate Program, Linguistics Program Studies, Udayana University Denpasar, Bali-Indonesia (P)

** STIMIK STIKOM-Bali, Renon, Depasar, Bali-Indonesia

EVALUATION OF THERMAL INDUSTRIAL AIR HEATERS BASED ON SOLAR SYSTEM

Y.Prasadkumar *

SK.Hyderali**

Abstract

The usage of solar energy heating systems increased greatly during the last decade. Need to make a setup which acquires much of the Sun's radiation as per the output requirement. More than one hundred years ago, black painted water tanks were used as simple solar water heaters in a number of countries. As the development takes place total setup gets advanced. Our main objective is to keep the system in such a way that, yields a better result. The major equipment includes collector, heat-exchanger, storage tank, and pumps & pipes.

Keywords:

Solar energy;
Solar water heaters;
Collector;
Heat-exchanger;
Storage tank.

Author correspondence:

First Author,

Y.Prasadkumar

M.Tech Student Department of *Mechanical Engineering*, Baba Institute of Technology and Sciences (BITS), Visakhapatnam (Dist.)

Second Author,

SK.Hyderali

M.Tech Student Department of *Mechanical Engineering*, Baba Institute of Technology and Sciences (BITS), Visakhapatnam (Dist.)

* Doctorate Program, Linguistics Program Studies, Udayana University Denpasar, Bali-Indonesia
** STIMIK STIKOM-Bali, Renon, Depasar, Bali-Indonesia

Effect of Fiber Orientation on Strength of a Cross-Ply Composite Plate using ANSYS

Reddipalli Srinivasarao*
Sheik .Hyderali**

Abstract

The objective of this paper is to Superior mechanical properties of composite materials such as high stiffness and strength to weight ratios, corrosive resistance and low coefficients of thermal expansion, caused it to be used increasingly in many areas of technology including marine, aerospace, automotive and others. To take advantage of full potential of composite materials, accurate models and design methods are required the most common structural elements are plates and shells. Laminated composites are one of the classifications of the composites which are used in structural elements like leaf springs, automobile drive shafts, and gears, and axles. An accurate understanding of their structural behaviour is required, such as the deflections, the through thickness distributions of stresses and strains, the large deflection behaviour and, of extreme importance for obtaining strong, reliable multi-layered structures. In general, the fibres in the composite materials can be of unidirectional and bidirectional. Here the focus is on unidirectional fibre composite materials. The fibres can be oriented in many ways i.e., 0, 10, 20, 30, 45, 90 degrees etc., but the strength in each type will be different. Using the load-deflection graphs, the maximum load and flexural strength are calculated. The attention is to note the variation of flexural strength of a composite material with the variation of fibre orientation and comparing the results with practical results. Here the modelling of composite material with different fibre orientations is done in Ansys software and simulation is done on them and those results are compared with practical results.

Keywords:

Flexural strength;
Laminated;
Fibre reinforced;
Glass fibre;
Graphite fibre;
Unidirectional;
Bidirectional;
Corrosion resistance.

Author correspondence:

First Author,
R Srinivasarao
M.Tech Student Department of Mechanical Engineering, Baba Institute of Technology and Sciences (BITS), Visakhapatnam (Dist.)
Second Author,
SK.Hyderali

* Doctorate Program, Linguistics Program Studies, Udayana University Denpasar, Bali-Indonesia (9 pt)

** STIMIK STIKOM-Bali, Renon, Denpasar, Bali-Indonesia



Fabrication and Experimental Investigation on Mechanical Properties of Aluminium Metal Matrix Composite by using Stir Casting

SK. Hyderali*

K. Satya Srikanth**

J. Pavani Kumar***

S. Muralidhar****

Abstract

From the last two decades there is a rapid increase in utilisation of Aluminium Alloy, particularly in Automobile, Aerospace Industries, due to its low density, low weight and high strength. Aluminium metal matrix composite has received extensive attention for practical as well as fundamental reasons. Aluminium Alloy & Aluminium based metal matrix composites (MMC) have found many applications in the manufacturing of various components. The present work is focused on Fabrication & Investigation of Mechanical properties of Al6061, Graphite, Fly ash metal matrix composite. These materials are identified in the process of making high strength Aluminium Alloy. Stir casting process is employed for the fabrication of metal matrix composite. In our project the composition of MMC is varied for graphite (3%, 6%) & fly ash (6%, 3%) by keeping 91% by Al 6061. Fabricated samples are tested for Mechanical properties. The Tests includes Tensile test, Hardness test, and Wear test as per ASTM standards. Microstructures of the samples were tested by using metallurgical microscope & compare the microstructure for each.

Keywords:

Composite materials;
Mechanical properties;
Metal matrix composites;
Stir casting;
Micro structure.

Author correspondence:

First Author,

SK. Hyderali

M.Tech Student Department of Mechanical Engineering, Baba Institute of Technology and Sciences (BITS), Visakhapatnam (Dist.)

* Doctorate Program, Linguistics Program Studies, Udayana University Denpasar, Bali-Indonesia (S p)

** STIMIK STIKOM-Bali, Renon, Denpasar, Bali-Indonesia

*** English Language Specialist, Offer Center, Carriage House, 2nd Floor, California, USA

**** Assistant Professor, Department of Mechanical Engineering, AITS, Kadapa, Andhra Pradesh, India



PRINCIPAL
Baba Institute of Technology and Sciences
Bakkannapalem (V), Madhurawada (P)
Visakhapatnam

Analysis of Bonded Tubular Single Lap Joint subjected to varying Pressure at constant Torsion

Vikas Ranjan,

Assistant Professor

Department of Mechanical Engineering, BITS-Vizag

vikas.ranjan7@gmail.com

R.R.Das

Associate professor

Department of mechanical Engineering, ISM- Dhanbad

Abstract: Fiber-reinforced polymer (FRP) composites are becoming increasingly popular in the engineering applications as alternative to conventional engineering materials. The unique characteristics of FRP, such as their light weight, their resistance to corrosion, high energy absorption, and the lower cost of transportation, erection and maintenance, are very promising in the application of FRP in various engineering fields. Bonded tubular structures made with FRP composites have been common structures in various fields of engineering. The purpose of this paper is to study the combined effect of internal pressure and torsion on stress distributions and failure within the joint region. Special attention has been devoted to study the effect of pressure increase at constant torsion on the failure prone regions.

Keywords: FRP, TSLJ, Composites, Pressure, Torsion, CCW

Introduction

Two Gr/E [0]₄ laminated FRP composite tubes which are similar with respect to length, thickness, and properties have been used as adherends. Here zero degree fiber orientation indicates circumferentially wound fibers. The two tubes have been joined through a thin layer of adhesive (epoxy) as shown in the Figure. The bonded TSLJ have been subjected to an internal pressure of 10 MPa at the inner adherend as well as torsion loading of 100 N-m (direction of the applied torque is CCW as we see from the free end of the bonded TSLJ) at the free end of the bonded TSLJ structure. Internal pressure loading has been varied as 1MPa, 10MPa, 13 MPa, and 16 MPa for a constant torsional loading of 100 N-m in order to study the effect of circumferential pressure variation on joint strength. The material properties along with strength values for adhesive and adherends have been given. The material properties have been considered from the work of Das and Pradhan (2010). Three different bondline interfaces have been identified to be the critical regions prone to stress concentration effects in the present analysis: (i) inner adherend-adhesive interface, (ii) adhesive mid-layer, and (iii) outer adherend-adhesive interface (Figure). Two-dimensional stress distribution comparison in different cases has been over viewed in this chapter.



Static and Vibrational Analysis on Automobile Composite Seat

G.Prasad

M.Tech CAD/CAM,

Baba Institute of Technology and Science.

Hepsiba Seeli

Assistant Professor,

Baba Institute of Technology and Science.

Abstract:

In Automobile, seat is one of the most important component of the vehicle, designing an optimal seat is always a challenging task for the engineers as the design parameters and comfort contains are always complex because it contains many parts and mechanisms for occupant safety and comfort. To design and develop an automobile seat with lesser cost and better occupants comfort is a prime principle for the engineer. Generally professional drivers spend eight to ten hours per a day, which will create a lot of pressure on the human tissue. Continuous or immobilized sitting with a contact posture for a long time will cause a lot of pressure and leads sub-dermal tissue damage. Tissue damage is occur because the total human weight is caring by it, which will causes a permanent deformation and ultimately rises the pressure on the human body. There are three main parameters in the seat design, Comfort, safety and health of the occupant.

This work accelerate and economize the development and design process of automobile seat and its reusability. Designing a seat which have moderate comfort and better vibrational stability is the objective of this work. The vibrations caused by the automobile and the uneven roads will effects the occupant health and its mode, these vibrations will create a lot of fatigue on the human body and will create stress concentration on critical zones on the human body. The modeling of the automobile seat is done in the CREO-2.0 parametric and the finite element analysis is conducted by using ANSYS 15.0. Polyurethane foams are used as the cushion and backrest materials for analysis. Higher density foam can absorb vibrations because of its closed packed Molecular structure, at the same time because of its closed packing arrangement it will be rigid in nature.

To achieve both vibrational stability and comfort ness composite seat is proposed where different materials are stacked one up on another so we get both comfortless and vibrational stability.

1. Introduction:

In today's international automobile market companies are driving the automobile characteristics sole for the customer choice and satisfaction. The working environment in the automobile is caricaturized as a confined space where the occupant interact with the automobile and adapt to perform the driving task. During the design and development of the car seat the occupant poster is very important, because it will create the comfort for driving engineers are said to be maximize the comfortless. The human perception of comfort will change through short and long period of operation. Seat means a structure which will support the occupant and provide the safety and comfort to the occupant.

Generally seat is mounted on a metal frame which is joined to the chassis or body of the automobile, when vibrations are generated due to the uneven road conditions where start at the tire of the automobile and it finally reaches to the driver. These vibrations will exerts forces on the human tissues and which will experiences a muscle contraction and final leads pain in long run. The material s used for are usually made form elastic material so that it will give greater comfort for the occupant but because of its elastic nature it will transmits the vibrational forces through it, which has to minimize for drivers safety. Materials which have better damping coefficient will absorb the vibrations better. Generally polyurethane were used as the material for seat, because of its elastic property and it's molecular structure.

FEASIBILITY STUDY ON NANO CRYOGENIC COOLANT TO BE USED IN FUTURISTIC HIGH TEMPERATURE SUPERCONDUCTING (HTS) CABLES

Rajesh Kumar Gadekula

Department of Mechanical Engineering, AITS, Kadapa, Andhra Pradesh, India.

Venkata Ramana Uppada

Department of Mechanical Engineering, BITS-Vizag, Andhra Pradesh, India.

Abhinav Kumar

School of Mechanical Engineering, LPU, Phagwara, Punjab, India.

Raja Sekhar Dondapati

Associate Professor, School of Mechanical Engineering, LPU, Phagwara, Punjab, India.

ABSTRACT

High Temperature Superconducting (HTS) Cables are predominantly used for efficient transmission of energy in power applications compared to the Conventional cables due to high current handling capacities with fewer transmission losses. In order to transmit the energy by superconducting power distribution and transmission systems, HTS cables are need to be cooled below cryogenic temperatures to retain its superconductivity. Hence, an appropriate cryogenic coolant should be identified. In this context, Nitrogen is used as a coolant because it is abundantly available in atmospheric air. The preset work aims at feasibility study on novel cooling of HTS cable by using nano cryogenic coolant. This coolant is prepared by dispersion of Copper Oxide (CuO) in Liquid Nitrogen (LN₂). Different mathematical models which are developed in the past used to investigate the thermophysical properties such as Specific heat, Thermal conductivity, Viscosity and Density. Properties are studied by varying volume concentration of CuO from 1-5% and temperature of LN₂ from 65-75 K for raise in every 1K. From the results, it is observed that dispersion of 5% CuO in LN₂ shows the heat transfer enhancement compared to LN₂. The obtained results shows nano cryogenic coolant can be used in futuristic HTS cables.

Key words: HTS cables, Nano cryogenic coolants, Liquid Nitrogen, Thermo physical properties, Heat transfer rate.

PRINCIPAL
Baba Institute of Technology and Sciences
Bakkannapalem (V), Madhurawada (P)
Visakhapatnam

editor@iaeme.com

A Survey on Rectifier for PE energy harvesting to maximize the potency transfer from the transducer to the load

MR .K. VIJAY¹, MR.PRADEEP KONDAPALLI ²& B.V.R.GOWRI³

¹PG Scholar Dept. of ECE in Baba Institute of Technology and Sciences,
Visakhapatnam, Andhra Pradesh, India.

²Associate professor Dept. of ECE in Baba Institute of Technology and Sciences,
Visakhapatnam, Andhra Pradesh, India.

³Assistant Professor Dept. of ECE in Baba Institute of Technology and Sciences,
Visakhapatnam, Andhra Pradesh, India.

ABSTRACT

This paper presents a self-powered rectifier for piezoelectric energy harvesting applications, and the key conception of the proposed system is to reset the transducer capacitor at optimal instants to maximize the extracted potency. The proposed rectifier consists of two switches and two active diodes. The switches discharge the transducer capacitor at optimal instants two times for every cycle. The active diodes are predicated on op-amps with a preset dc offset, which reduces the voltage drop and the leakage current and eschews instability. In advisement, the controller for the proposed rectifier is simple to reduce the circuit involution and the potency dissipation. The proposed rectifier was designed and fabricated in 0.18- μm CMOS technology. Quantified results designate that it achieves high power efficiency and the amount of potency extracted by the proposed rectifier is more astronomically immense when compared with the conventional rectifiers. The proposed rectifier does not require any off chip components to enable a full chip integration.

Key words: - Active Rectifier, High Efficiency Rectifier, op-Amp-Based Active-Diodes, Piezoelectric Cantilever, Piezoelectric Energy Harvesting, Synchronized Switch Harvesting On Inductor ,Piezo Electric, Energy Harvesting, Dc Offset, Symmetrical Slew Rate, Class-Ab Op-Amp, Self-Powered Rectifier.

1. INTRODUCTION

In the recent past years many researchers have been working on the energy harvesting. Abstraction of batteries is very utilizable in many applications such as wireless sensor

networks where transmuting of batteries is arduous and costly. Energy harvesting has given a solution for these applications where supersession of batteries is impractical. Piezoelectric energy (PE) harvesting

A Survey on Rectifier for PE energy harvesting to maximize the potency transfer from the transducer to the load

MR .K. VIJAY¹, MR.PRADEEP KONDAPALLI ²& B.V.R.GOWRI³

¹PG Scholar Dept. of ECE in Baba Institute of Technology and Sciences,
Visakhapatnam, Andhra Pradesh, India.

²Associate professor Dept. of ECE in Baba Institute of Technology and Sciences,
Visakhapatnam, Andhra Pradesh, India.

³Assistant Professor Dept. of ECE in Baba Institute of Technology and Sciences,
Visakhapatnam, Andhra Pradesh, India.

ABSTRACT

This paper presents a self-powered rectifier for piezoelectric energy harvesting applications, and the key conception of the proposed system is to reset the transducer capacitor at optimal instants to maximize the extracted potency. The proposed rectifier consists of two switches and two active diodes. The switches discharge the transducer capacitor at optimal instants two times for every cycle. The active diodes are predicated on op-amps with a preset dc offset, which reduces the voltage drop and the leakage current and eschews instability. In advisement, the controller for the proposed rectifier is simple to reduce the circuit involution and the potency dissipation. The proposed rectifier was designed and fabricated in 0.18- μm CMOS technology. Quantified results designate that it achieves high power efficiency and the amount of potency extracted by the proposed rectifier is more astronomically immense when compared with the conventional rectifiers. The proposed rectifier does not require any off chip components to enable a full chip integration.

Key words: - Active Rectifier, High Efficiency Rectifier, op-Amp-Based Active-Diodes, Piezoelectric Cantilever, Piezoelectric Energy Harvesting, Synchronized Switch Harvesting On Inductor ,Piezo Electric, Energy Harvesting, Dc Offset, Symmetrical Slew Rate, Class-Ab Op-Amp, Self-Powered Rectifier.

1. INTRODUCTION

In the recent past years many researchers have been working on the energy harvesting. Abstraction of batteries is very utilizable in many applications such as wireless sensor

networks where transmuting of batteries is arduous and costly. Energy harvesting has given a solution for these applications where supersession of batteries is impractical. Piezoelectric energy (PE) harvesting

Frequency Reconfigurable Log Periodic Microstrip Dipole Antenna Array for Wideband Applications

B. Siva Prasad

Associate Professor

*Department of Electronics & Communication Engineering
Baba Institute of Technology & Sciences, Visakhapatnam,
Andhra Pradesh*

M. G. Sumanth

*Department of Electronics & Communication Engineering
A.U. College of Engineering, Andhra University,
Visakhapatnam, Andhra Pradesh*

Prof. P. Mallikarjuna Rao

*Department of Electronics & Communication Engineering
A.U. College of Engineering, Andhra University, Visakhapatnam, Andhra Pradesh*

Abstract

The concept of reconfigurable antenna is widely used as an additional feature of reconfigurable ability for future wireless communication systems especially for cognitive radio applications, radar system and multi-frequency communications. There are various configurations of reconfigurable antenna such as monopole, dipole and log-periodic wideband antenna. This paper describes the design and simulation of frequency reconfigurable log periodic microstrip dipole array (FRLPMDA) using dipole patch elements with an useful frequency range for C band (4-8 GHz), X band (8-12 GHz) and Ku band (12-18 GHz). The integration of reconfigurable antennas with radio frequency (RF) switches are needed to perform the switchable ability [3-6]. In this paper, a frequency reconfigurable LPMDA has been designed to perform a wideband frequency operations by connecting fifteen dipole antennas using coaxial feed line technique. Then, the diversity in frequency is achieved by switching ON and OFF of the PIN diodes connected at every transmission line. It is compact in size and useful to support on board Radar and many wireless applications where they can reduce the size of front circuitry in communication systems.

Keywords: Reconfigurable, Log Periodic Antenna, Microstrip, Wideband

I. INTRODUCTION

Design of the modern microwave communication systems and on board radar equipment for air craft and war ships EW/ECM needs a proper technical solution for small size broadband antennas. The main goal of designing this antenna is to obtain frequency diversity. A single antenna can be operated over three different sub bands. Suppose an enemy is jamming our present frequency band with noise signal, then we can shift our present operating frequency band by using same antenna as before, without changing its physical design by controlling of the switches at the transmission line of patch. The required frequency band could be achieved for three sub bands by controlling a group of switches as shown in table 2. For this purpose, reconfigurable log periodic microstrip dipole array is used for broad operating bandwidth of 14 GHz. Here ANSYS HFSS 13.0 v is used to carry out the simulation process. The antenna is analyzed based on several parameters such as Return loss, Gain and VSWR. It is a low cost multiband printed circuit board (PCB) antenna and it can be fabricated on Arlon AD 250 substrate with Relative Dielectric Constant $\epsilon_r = 2.5$ and thickness $h = 0.51\text{mm}$.

II. ANTENNA DESIGN

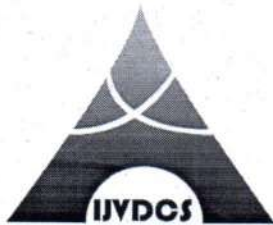
The standard approach for the design of LPDAs proposed by Carrel[1] can be used with suitable modifications. The geometrical structure of the proposed frequency reconfigurable log periodic microstrip dipole array antenna is shown in Figure.1 with fifteen dipole patches which are developed on Arlon AD 250 substrate with thickness of 0.51mm and dimensions of 70×40 mm. The design principle of log periodic microstrip dipole array antenna requires scaling of dimensions from period to period so that the performance is periodic with logarithm of frequency. Scaling of dimensions of patch dipole elements from one dipole element to another is given by scaling factor ' τ ' which determines the antenna angle apex ' α '. spacing factor ' σ ' which is also known as "Relative Spacing constant". Patch length (L_n), width (W_n) and distance between elements ($d_{n,n+1}$) are calculated [2] and they are presented in table 1.

For this design $f_h = 18\text{ GHz}$, $f_l = 4\text{ GHz}$.

1) STEP 1: The Scale Factor $\tau = 0.850$ and
Spacing Factor $\sigma = 0.14$


PRINCIPAL

**Baba Institute of Technology and Sciences
Bakkannapalem (V), Madhurawada (P)
Visakhapatnam**



www.ijvdc.org

International Journal of VLSI System Design and Communication Systems

ISSN 2322-0929
Vol.05, Issue.08,
August-2017,
Pages:0786-0790

VLSI Realization of Hamming Logic Based Error Corrected Parallel Filters

SURAPATHI RAMU¹, K.KALAPRIYA²

¹PG Scholar, Baba Institute of Technology and Sciences, P.M.Palem, Vishakhapatnam, Andhra Pradesh, India,
Email: sramu.5713@gmail.com.

²Assistant Professor, Baba Institute of Technology and Sciences, P.M.Palem, Vishakhapatnam, Andhra Pradesh, India,
Email: kostikalapriya@gmail.com.

Abstract: Digital filters are widely used in signal processing and communication systems. In some cases, the reliability of those systems is critical, and fault tolerant filter implementations are needed. Over the years, many techniques that exploit the filters' structure and properties to achieve fault tolerance have been proposed. As technology scales, it enables more complex systems that incorporate many filters. In those complex systems, it is common that some of the filters operate in parallel, for example, by applying the same filter to different input signals. Recently, a simple technique that exploits the presence of parallel filters to achieve fault tolerance has been presented. In this brief, that idea is generalized to show that parallel filters can be protected using error correction codes (ECCs) in which each filter is the equivalent of a bit in a traditional ECC. This new scheme allows more efficient protection when the number of parallel filters is large. The technique is evaluated using a case study of parallel finite impulse response filters showing the effectiveness in terms of protection and implementation cost.

Keywords: Parallel filters, Error Correction Codes (ECCs), Reversible Logic Gates, Verilog.

I. INTRODUCTION

Electronic circuits are increasingly present in the automotive, medical, and space applications where reliability is critical. In those applications, circuits have to provide some of fault tolerance of degree. This need is further increased by the intrinsic reliability challenges of advanced technologies of CMOS that include, e.g., manufacturing variations and soft of errors. A number of the techniques can be used to protect a circuit from the errors. Those range from modifications in the manufacturing process of the circuits to reduce the number of errors to adding redundancy at the logic or system level to ensure that the errors do not affect of the system functionality. To add redundancy, a general technique known as a triple modular redundancy (TMR) can be used. The TMR, which triplicates the design and adds voting logic to correct errors, is commonly used. However, it more than the triples area and power of the circuit, something that may not be acceptable in some of the applications. When the circuit to be protected has algorithmic or structural of the properties, a better option can be to exploit those properties to implement of the fault tolerance. One example is signal processing circuits for which the specific some techniques have been proposed over the years. Digital filters are one of the most commonly used in the signal processing circuits and several techniques have been proposed to protect them from errors. For example, the use of reduced precision replicas was proposed to reduce the cost of implementing modular redundancy in FIR filters.

Other schemes have exploited the FIR properties at a word level to also achieve fault tolerance. A relationship between the memory elements of an FIR filter and the sequence of input was used to detect errors. We consider where two parallel filters with the same response that processed different input signals. However, it is increasingly common to find systems in which the several filters operate in parallel. This is the case in filter banks and in many of the modern communication systems. For those systems, the protection of filters can be addressed at a higher level by considering the parallel filters as the block to be protected. This idea was explored. It was shown that with only copy of the one redundant, single error correction can implemented. Therefore, a significant reduction of cost compared with TMR was obtained. The use of number systems of residue and arithmetic codes has also been proposed to protect filters. Finally, the FIR filters are used to different implementation structures to correct errors with only one of the redundant module has also been proposed. In all the techniques mentioned so far, the protection of a single filter is considered. In this brief, a general scheme to protect parallel filters is presented. Parallel filters with the same response that process of different input signals are considered. The new approach is based on the application of error correction codes (ECCs) using each outputs of the filter as the equivalent of bit in and ECC codeword. This is a generalization of the scheme is presented and enables the more efficient implementations when the number of parallel filters is large not only small. The scheme can also be used to

Operational and Productivity Performance of Andhra Pradesh State Financial Corporation: An Experimental Analysis

Dr. P. Sanjeevi,

*Professor, Department of Management Studies,
BITS VSP*

Mr. M. Vasudeva Rao,

*Assistant Professor, DMS,
BITS VSP*

MR. S. Souman Dutta,

*Assistant Professor, DMS,
BITS VSP*

Abstract

Indian banking industry has transformed in recent years due to globalization in the world market, which has resulted in fierce competition. In this paper, an attempt has been made to analyze the operational performance and productivity performance of Andhra Pradesh State Financial Corporation. Analytical frame of the operational aspects in terms of the sanctions, disbursements, collection, operating profit, number of accounts and number of employees and in general, and to different sectors, purposes; and also, the trends over the five years period under study have been presented. In addition, the operational and productivity performance is evaluated by the known parameters and bench marks. The study is based on secondary data collected from annual reports of APSEC for the study period i.e. 2013-14 to 2017-18. Accounting Ratios, t-test, correlation analysis has been used to analyze the present data. The study suggested that APSEC should highlight on generating more profits by efficient utilization of its assets, capital and improving the productive efficiency of number of accounts and employees.

Key words: Productivity Performance, Operational Performance, Sanctions, Disbursement, collection, operating profit.

INTRODUCTION

The mandate given in the statute drives the corporation to develop entrepreneurship, bring in balanced regional development and drive the industrialization in the state in order to bring the economic

Operational and Productivity Performance of Andhra Pradesh State Financial Corporation: An Experimental Analysis

Dr. P. Sanjeevi,

*Professor, Department of Management Studies,
BITS, VSP*

Mr. M. Vasudeva Rao,

*Assistant Professor, DMS,
BITS, VSP*

MR. S. Suban Dutta,

*Assistant Professor, DMS,
BITS, VSP*

Abstract

Indian banking industry has transformed in recent years due to globalization in the world market, which has resulted in fierce competition. In this paper, an attempt has been made to analyze the operational performance and productivity performance of Andhra Pradesh State Financial Corporation. Analytical frame of the operational aspects in terms of the sanctions, disbursements, collection, operating profit, number of accounts and number of employees and in general, and to different sectors, purposes; and also, the trends over the five years period under study have been presented. In addition, the operational and productivity performance is evaluated by the known parameters and bench marks. The study is based on secondary data collected from annual reports of APSEC for the study period i.e. 2013-14 to 2017-18. Accounting Ratios, t-test, correlation analysis has been used to analyze the present data. The study suggested that APSEC should highlight on generating more profits by efficient utilization of its assets, capital and improving the productive efficiency of number of accounts and employees.

Key words: Productivity Performance, Operational Performance, Sanctions, Disbursement, collection, operating profit.

INTRODUCTION

The mandate given in the statute drives the corporation to develop entrepreneurship, bring in balanced regional development and drive the industrialization in the state in order to bring the economic

Operational and Productivity Performance of Andhra Pradesh State Financial Corporation: An Experimental Analysis

Dr. P. Sanjeevi,

*Professor, Department of Management Studies,
BITS VSP*

Mr. M. Vasudeva Rao,

*Assistant Professor, DMS,
BITS VSP*

MR. S. Seeman Dutta,

*Assistant Professor, DMS,
BITS VSP*

Abstract

Indian banking industry has transformed in recent years due to globalization in the world market, which has resulted in fierce competition. In this paper, an attempt has been made to analyze the operational performance and productivity performance of Andhra Pradesh State Financial Corporation. Analytical frame of the operational aspects in terms of the sanctions, disbursements, collection, operating profit, number of accounts and number of employees and in general, and to different sectors, purposes; and also, the trends over the five years period under study have been presented. In addition, the operational and productivity performance is evaluated by the known parameters and bench marks. The study is based on secondary data collected from annual reports of APSEC for the study period i.e. 2013-14 to 2017-18. Accounting Ratios, t-test, correlation analysis has been used to analyze the present data. The study suggested that APSEC should highlight on generating more profits by efficient utilization of its assets, capital and improving the productive efficiency of number of accounts and employees.

Key words: Productivity Performance, Operational Performance, Sanctions, Disbursement, collection, operating profit.

INTRODUCTION

The mandate given in the statute drives the corporation to develop entrepreneurship, bring in balanced regional development and drive the industrialization in the state in order to bring the economic

CONSUMER ATTITUDE TOWARDS THE PRODUCTS OF HYPER MARKETS IN VISAKHAPATNAM

Dr. B. Rama Jyothi*

Mr. M. Vasudeva Rao**

Mr. Naresh Kumar Kuppli***

Abstract

Market is a place where buyers and sellers meet, goods and services are offered for sale and transfer of ownership occurs. Sometimes a market is a precisely physical place and sometimes it refers to groups of people. At present day's hyper markets plays an important role to develop the modern markets and improve the customer services and promote the economic excelleration. As market consists of consumers, their attitude, beliefs, loyalty and perception is important in modern marketing. Without customers, a firm has no revenues, no profits, and therefore no market value. Consumers all have different tastes, like and dislikes and adopt different behaviour patterns while making purchase decisions. The present study deals with the attitude, consumer satisfaction and preferences with regard to the services, prices, quality, products etc. of hyper markets.

Keywords:

Modern Markets;
Customer Attitude;
Customer satisfaction;
Hyper markets;

Copyright © 201x International Journals of Multidisciplinary Research Academy. All rights reserved.

Author correspondence:

Dr. B. Rama Jyothi,
Assistant Professor,
Baba Institute of Technology and Sciences,
PM palem,
Visakhaptnam-48.
Email: bramajyothi@bitsvizag.com

Mr. M. Vasudeva Rao**
Assistant Professor,
Baba Institute of Technology and Sciences,
PM palem,
Visakhaptnam-48.
Email: vasumalicherla@gmail.com

Mr. Naresh Kumar Kuppli***
Assistant Professor,
Baba Institute of Technology and Sciences,
PM palem,
Visakhaptnam-48.
Email: nareshpatnaik.k@gmail.com

* Doctorate Program, Linguistics Program Studies, Udayana University Denpasar, Bali-Indonesia (9 pt)
** STIMIK STIKOM-Bali, Renon, Denpasar, Bali-Indonesia

International Journal of Engineering, Science and Mathematics

Vol. x issue x, Month 201x,

ISSN: 2320-0294 Impact Factor: 6.765

Journal Homepage: <http://www.ijesmj.co.in>, Email: ijesmj@gmail.com

Double-Blind

Peer Reviewed Refereed Open Access International Journal - Included in the International Serial Directories Indexed & Listed at Ulrich's Periodicals Directory ©, U.S.A., Open J-Gate as well as in Cabell's Directories of Publishing Opportunities, U.S.A.

CONSUMER ATTITUDE TOWARDS THE PRODUCTS OF HYPER MARKETS IN VISAKHAPATNAM

Dr. B. Rama Jyothi*

Mr. M. Vasudeva Rao**

Mr. Naresh Kumar Kuppli***

Abstract

Market is a place where buyers and sellers meet, goods and services are offered for sale and transfer of ownership occurs. Sometimes a market is a precisely physical place and sometimes it refers to groups of people. At present day's hyper markets plays an important role to develop the modern markets and improve the customer services and promote the economic excelleration. As market consists of consumers, their attitude, beliefs, loyalty and perception is important in modern marketing. Without customers, a firm has no revenues, no profits, and therefore no market value. Consumers all have different tastes, like and dislikes and adopt different behaviour patterns while making purchase decisions. The present study deals with the attitude, consumer satisfaction and preferences with regard to the services, prices, quality, products etc. of hyper markets.

Keywords:

*Modern Markets;
Customer Attitude;
Customer satisfaction;
Hyper markets;*

Copyright © 201x International Journals of Multidisciplinary Research Academy. All rights reserved.

Author correspondence:

Dr. B. Rama Jyothi,
Assistant Professor,
Baba Institute of Technology and Sciences,
PM palem,
Visakhapatnam-48.
Email: bramajyothi@bitsvizag.com

Mr. M. Vasudeva Rao**
Assistant Professor,
Baba Institute of Technology and Sciences,
PM palem,
Visakhapatnam-48.
Email: vasumalicherla@gmail.com

Mr. Naresh Kumar Kuppli***
Assistant Professor,
Baba Institute of Technology and Sciences,
PM palem,
Visakhapatnam-48.
Email: nareshpatnaik.k@gmail.com

* Doctorate Program, Linguistics Program Studies, Udayana University Denpasar, Bali-Indonesia (9 pt)

** STIMIK STIKOM-Bali, Renon, Depasar, Bali-Indonesia

RURAL WOMEN ENTREPRENEURS ROLE &CHANGE IN THE SOCIETY: CASE STUDY

Dr. SREENIVASA RAO BEHARA

Professor, Department of Management Studies,
BABA Institute of Technology and Sciences (BITS),
PM Palem, Visakhapatnam- 500 048

MR P.B. NARENDRA KIRAN

Head, Department of Management Studies,
BABA Institute of Technology and Sciences (BITS),
PM Palem, Visakhapatnam- 500 048

Abstract:

A society that does not have an optimistic, positive empowering image of the future is endangered. When society is skewed & power, it begins an entropic spiral to isolation – Entrepreneurs can their skills to restructure the society. Society entrepreneurs are change agents that improve society by new models & developing equal effective often less hierarchical complex than existing ones. Society entrepreneurship mainly defined by Researchers and practitioners in various ways with common factor that adds value to a community mission through innovative, risk-taking, similar business practices. Society entrepreneurship combines innovation with the importance of community needs and their work overlaps with social justice and environmental presentation movements. Ancient Indian Women were House Wives and were accorded to follow their husbands in their religious duties. Gradually the Change came into varying social-political balance, women were changed their background aggressively dramatic exploitation.

Keywords: empowerment, entrepreneur & innovation.


PRINCIPAL
Baba Institute of Technology and Sciences
Bakkannapalem (V), Madhurawada (P)
Visakhapatnam

RURAL WOMEN ENTREPRENEURS ROLE & CHANGE IN THE SOCIETY: CASE STUDY

Dr. SREENIVASA RAO BEHARA

Professor, Department of Management Studies,
BABA Institute of Technology and Sciences (BITS),
PM Palem, Visakhapatnam- 500 048

MR P.B. NARENDRA KIRAN

Head, Department of Management Studies,
BABA Institute of Technology and Sciences (BITS),
PM Palem, Visakhapatnam- 500 048

Abstract:

A society that does not have an optimistic, positive empowering image of the future is endangered. When society is skewed & power, it begins an entropic spiral to isolation – Entrepreneurs can their skills to restructure the society. Society entrepreneurs are change agents that improve society by new models & developing equal effective often less hierarchical complex than existing ones. Society entrepreneurship mainly defined by Researchers and practitioners in various ways with common factor that adds value to a community mission through innovative, risk-taking, similar business practices. Society entrepreneurship combines innovation with the importance of community needs and their work overlaps with social justice and environmental presentation movements. Ancient Indian Women were House Wives and were accorded to follow their husbands in their religious duties. Gradually the Change came into varying social-political balance, women were changed their background aggressively dramatic exploitation.

Keywords: empowerment, entrepreneur & innovation.


PRINCIPAL
Baba Institute of Technology and Sciences
Bakkannapalem (V), Madhurawada (P)
Visakhapatnam



TRAINING AND DEVELOPMENT PRACTICES IN SELECT CORPORATE RETAIL STORES IN VISAKHAPATNAM

Mr. P.BalaNarendraKiran

Head, Department of Management Studies
Baba Institute of Technology and Sciences,
Visakhapatnam, Andhrapradesh

K.Sireesha

Assistant Professor,
Department of Management Studies
Baba Institute of Technology and Sciences,
Visakhapatnam, Andhrapradesh

Abstract:

A profound training program acts as a vehicle to enhance employee skills and enable them to perform better in their job. Training and development is very crucial to the employees, the organization and their effectiveness. Training is important in retailing because majority of the employees have direct contact with customers. They are responsible for helping customers, satisfy their needs and resolve their problems. Training is important not only from the point of view of employees, but also for the organization. This study has examined the perception of respondents on the present training and development practices of select corporate retail organizations. With this background, this study is conducted to know the perception of respondents in terms of their demographic profiles of gender, age, education and social status.

Keywords: Training, Development, Promotion

1. Introduction

Training programmes are created not only for the new recruits but also often designed for the existing personnel within the organization especially when they are either transferred or promoted. All large and small retail organizations must conduct some form of a training programme to be able to perform smoothly. This is a part of the overall induction programme that is exclusively designed for every position. A comprehensive training and development program helps "in deliberating on the knowledge, skills and attitudes necessary

TRAINING AND DEVELOPMENT PRACTICES IN SELECT CORPORATE RETAIL STORES IN VISAKHAPATNAM

Mr. P.BalaNarendraKiran

Head, Department of Management Studies
Baba Institute of Technology and Sciences,
Visakhapatnam, Andhrapradesh

K.Sireesha

Assistant Professor,
Department of Management Studies
Baba Institute of Technology and Sciences,
Visakhapatnam, Andhrapradesh

Abstract:

A profound training program acts as a vehicle to enhance employee skills and enable them to perform better in their job. Training and development is very crucial to the employees, the organization and their effectiveness. Training is important in retailing because majority of the employees have direct contact with customers. They are responsible for helping customers, satisfy their needs and resolve their problems. Training is important not only from the point of view of employees, but also for the organization. This study has examined the perception of respondents on the present training and development practices of select corporate retail organizations. With this background, this study is conducted to know the perception of respondents in terms of their demographic profiles of gender, age, education and social status.

Keywords: Training, Development, Promotion

1. Introduction

Training programmes are created not only for the new recruits but also often designed for the existing personnel within the organization especially when they are either transferred or promoted. All large and small retail organizations must conduct some form of a training programme to be able to perform smoothly. This is a part of the overall induction programme that is exclusively designed for every position. A comprehensive training and development program helps “in deliberating on the knowledge, skills and attitudes necessary

A Geo-Spatial approach to the Land Use/Land Cover mapping of the centre part of Srikakulam district, Andhra Pradesh, India

B. Sridhar

Department of Geo-Engineering

Andhra University College of Engineering (A) Visakhapatnam, Andhra Pradesh, India
sridhar.bendalam@gmail.com, Tel.: +91-8341556390

P Jagadeeswara Rao

Department of Geo-Engineering

Andhra University College of Engineering (A) Visakhapatnam, Andhra Pradesh, India
Neela Victor Babu

Baba Institute of Technology and Science, Visakhapatnam

Aditya Allamraju

Department of Geo-Engineering

Andhra University College of Engineering (A) Visakhapatnam, Andhra Pradesh, India

Abstract-In the present investigation, the land use/land cover mapping of the centre part of Srikakulam district in the state of Andhra Pradesh has been carried out using Resourcesat-1, LISS IV satellite imagery based on standard visual interpretation techniques. The identified land use / land cover features are Aquaculture, Coastal sand, Coastal wetland, Mud-flat, Coastal-wetland, Creek, Deciduous dense forest, Deciduous open forest, Forest-plantations, Industrial, Kharif (single crop), Kharif +Rabi(double-cropped), Plantation, Tank, Towns/cities(Urban), Villages(Rural), Wastelands, Barren-rock, Wastelands, Gullied land, Wastelands, hills with dense vegetation, Wastelands-hills with scrub, Land without-scrub, Salt affected land and Stone-waste. Out of all these features, the maximum area is covered by Kharif +Rabi (double-cropped) (28.091%) followed by Kharif (single crop) (22.773%), Plantation (13.13 %) and Wastelands, hills with scrub (10.916%).

Keywords: Land Use, Land Cover, Resourcesat-1, LISS IV, satellite imagery, Visual Interpretation

1. INTRODUCTION

Land and water have been the basic elements of life supporting systems on our planet since the dawn of civilization (Redclift, M. 2002). All great civilizations flourished where these resources were available in plenty and they declined or perished with the depletion of these resources. In recent years, the land resource has been subjected to a variety of pressures, yet it is surviving and sustaining mankind. The concerned matter is the way, land is being overexploited which is leading to threat to the environment. Out of all the species on the earth, man is the chief culprit to this degradation. He views land in terms of its utility to meet his perceived needs and wants. The most easily categorized varieties of land from the utility point of view are land fit for use, land with potential for use and land which appear useless in the foreseeable future.

Here probably lies the genesis of the problem of land degradation and erosion of ecosystems. Mahatma Gandhi said - "The Earth has enough for everybody's need but not for anybody's greed". Preserving, protecting and defending the land resources have been part of our age-old culture. The respect for the importance of land resources is best depicted in the conventional concept of Panchabhutas - land, water, fire, sky and air that

Baba Institute of Technology and Sciences
Bakkannapalem (V), Madhurawada (P)
Visakhapatnam

Geo-spatial analysis of HIV Status in East Godavari District, Andhra Pradesh, India

Dr.K Anand RatnaKumar
Dept.of Geography
Andhra University, Visakhapatnam - 530 003

Suribabu Boyidi
Senior Research Fellow
Dept of Geo Engg AUCE(A)- Andhra University

Dr.N Victor Babu
Professor &HOD,
Dept of Civil, BITS Vishakapatnam, .

Abstract: East Godavari district is one of the hardest hit states of HIV of Andhra Pradesh State in Southern India. Most part of the district is infested by HIV incidence. The year-wise data on HIV disease collected from the records of District Leprosy Office, Kakinada for the year 2002-2008 and cumulative childrens household's HIV cases for fourteen mandals of the district were collected from St.Paul's Trust, Samalkot from July, 2007 to March, 2009. Analysis of the data indicated that a total of 8561 cases were identified in the district. The analysis of HIV disease reveals that the number of HIV positive cases are steadily increasing since 2002-2003. The number was doubled during 2004-2005 and became more than 8000 by 2007-2008. Data is not available most part of the district and are available for fourteen mandals namely Samalkot, Biccavolu, Anaparthi, Mandapeta, Peddapuram, Rangampeta, Rajanagaram, Jaggampeta, Candepalle, Kakinada (Rural), Kakinada (Urban), Karapa, Pedapudi and Rayavaram. Among the mandals Peddapuram, Samalkot, Jaggampeta, Biccavolu and Mandapeta are highly prevalence of the disease low incidence in the mandals namely Rangampeta, Rayavaram and Karapa. Peddapuram records the high incidence of the disease with 1411 cases and followed by Samalkot and Jaggampeta. The maximum deaths were recorded in these mandals namely Peddapuram, Biccavolu and Rayavaram. Peddapuram records the high mortality with 35 and followed by Biccavolu and Rayavaram and less number of deaths recorded at Rangampeta. In the present study the spatial patterns of HIV incidence and mortality were made.

Introduction

Health is a state of complete, physical, mental and social well-being and not merely an absence of disease (WHO report 1948). The term disease is associated with health hazard that literally means 'without ease' (dis-ease). Park and Park (1977) defined disease as a condition in which body health is impaired, an alteration of the human body interrupting the performance of vital functions.

Acquired Immunodeficiency Syndrome (AIDS) represents the late clinical stage of infection with the Human Immunodeficiency Retro Viruses (HIV). The virus causing AIDS was first identified and called as 'lymphadenopathy-associated virus' (LAV) by a French scientists. United States researchers named it as 'human T-cell lymphotropic virus -III (HTLV-III). In May 1986, the International Committee on Taxonomy gave it new name as human immunodeficiency virus (HIV) (Anupa Mukhopadhyay et al., 1996). It has emerged as leading morbidity and premature mortality in cities around the world in a short period of over a decade. In North America it had become the principal killer of young adult males and it has outstripped overdose, suicide, homicide and hepatitis B (Anupa Mukhopadhyay et al, 2004) AIDS most often results in progressive damage to the immune, organ and central nervous systems.

Acquired Immuno Deficiency Syndrome A - Acquired (something that you get) I - Immune (capability of the human body to fight diseases)

PRINCIPAL
Baba Institute of Technology and Sciences
Bakkannapalem (V), Madhurawada (P)
Visakhapatnam

Page No:5919

A STUDY ON IDENTIFICATION OF LANDSLIDE ZONES ALONG THE RECENT EXPANSION OF GVMC, VISAKHAPATNAM DISTRICT-A GIS APPROACH

Neela Victor Babu

*Associative Professor, Department of Civil Engineering
Baba Institute of Technology and Science, Visakhapatnam
neelavictorbabu@gmail.com*

Ramojirao Jogi

*Department of Geo-Engineering, College of Engineering (A),
Andhra University, Visakhapatnam*

Gudikandhula Narasimha Rao

*Department of Geo-Engineering, College of Engineering (A),
Andhra University, Visakhapatnam*

Venkateswarlu Sunkari

*Assistant Professor, Center of ITSC, Addis Ababa Institute of technology,
Addis Ababa University, Addis Ababa, Ethiopia*

Abstract: The Greater Visakhapatnam Municipal Corporation (GVMC) city forms the area of the study in the present investigation. The population in the city is increasing at a rapid pace due to migration from surrounding areas, with corresponding increase in sub-urban areas also. The extensive growth of residential apartments along the coastal stretch resulted in excess withdrawal of groundwater, initiating seawater intrusion in these tracts. All these factors combined, leading to an acute environmental crisis in the area. Forecasting urban water demand can be of use in the management of water utilities (Salvatore Campisi-Pinto, 2012). Hence, it is necessary to manage the available water resources in an efficient manner. This is done using advanced technologies like Remote Sensing and Geographical Information System. The sites for water harvesting structures and techniques like check dams, percolation ponds, infiltration wells, afforestation, contour trenching etc.. previous literature gives us the techniques of Image Interpretation, concepts of remote sensing which help in identifying different features in the study area. Papers related to Geographic Information System tell us about the concepts of GIS which help in overlaying different layers for selecting the harvesting structures.

Keywords: LANDSLIDE ZONES, GVMC, R.S and GIS, INVESTIGATION

I. INTRODUCTION

Water being one of the most important natural resources need to be conserved and managed properly. As water covers 71 percent of the Earth's surface only three percent is fresh water and the rest 97 percent of the water on the Earth is salt water. 0.687 percent of total water resources are available as

PRINCIPAL

**Baba Institute of Technology and Sciences
Bakkannapalem (V), Madhurawada (P)
Visakhapatnam**

Application of Geospatial Technologies in Geology mapping of Bheemili in the district of Visakhapatnam, Andhra Pradesh, India

B. Sridhar

Department of Geo-Engineering, Andhra University College of Engineering (A), Visakhapatnam
sridhar.bendalam@gmail.com, Tel.: +91-8341556390

Neela Victor Babu

Baba Institute of Technology and Science, Visakhapatnam

Aditya Allamraju

Abstract- A geological map gives primary data for analyzing both past and present day to day processes affecting a region on the Earth. This kind of information is vital for providing information relating to geology that can aid to minimize damage and death caused by geologic hazards such as earthquakes and landslides. In the present investigation, thematic maps of geology have been produced from the IRS-ID-LISS III, 2004 and standard visual interpretation methods have been followed to portray on-screen digitations of the features. This study shows that three major kinds of rocks are exposed in the area, out of which khondalite is in majority followed by Quartzite and then Charnockite. The structural trend of rocks is NNE-SSW considering the local variations at few places. These features play an important role with respect to drainage development and erosion in the area.

Keywords: Geological hazards, Khondalite, Quartzite, Charnockite, Erosion

1. INTRODUCTION

GIS is a new and effective tool in geological mapping that helps the surveyors to generate 3-D maps of any area with accurate and desired scaling. In the present analysis, the geological thematic layers have been studied to identify the geological vulnerable areas along the road management. Standard on-screen interpretation techniques as per the nomenclature proposed by NRSA, have been followed and delineated the features on-screen. These values are obtained to understand the rock and soil behavior in sub-aerial erosions. The road constructed along the sea shore, thus the influence of marine activity on the road is also explored to locate areas vulnerable under marine conditions. The rocks along the coast are observed as khondalite, leptynite that belong to the metamorphic group. In general, the trend coincides with the trend of Eastern Ghats. The khondalite appears red to brown color rich in sillimanite, garnets and occasional occurrence of graphite, biotitic, etc. in subordinate amount. Huge red kanker or talus material has been observed along the road cutting. This material is liable to slide on to the road. Bad land topography is famous in the area. Highly gullied land masses with sparse vegetation, red soils, and quick sands are the hallmarks of the area. A number of non-perennial streams erode the area into bad land topography.


PRINCIPAL
Baba Institute of Technology and Sciences
Bakkannapalem (V), Madhuravada (P)
Visakhapatnam

STUDY ON M25 AND M30 GRADE CONCRETE BY PARTIAL REPLACEMENT OF CEMENT WITH METAKAOLIN

Ch.Kameswara Rao

Asst.Prof, Pydah College of Engineering & Technology

Dr.N.Victor Babu

Professor & H.O.D., Dept. of Civil Engg.,

BABA Institute of Technology and Sciences-Visakhapatnam

M.Sumesh Raja

Asst.Prof, Pydah College of Engineering & Technology

Abstract- Now a days the partially replacement of Ordinary Portland cement with pozzolana has been increasing widely. Most commonly used pozzolana are fly ash, silica fume, metakaolin, and ground granulated blast furnace slag. A pozzolana is a material which, when combined with calcium, exhibits cementitious properties. Pozzolana are commonly used as an addition to cement concrete mixture to increase the long-term strength and other material properties of cement concrete and in some cases to reduce the material cost of concrete. Recent researches aimed at energy conservation in the cement and concrete industry has in part, focused on the use of less energy intensive materials such as fly ash, slag and silica fume. Of the some attention has been given to the use of natural pozzolana like Metakaolin as possible partial replacement for cement. Amongst the various methods used to improve the strength, Durability and resisting power of the temperature, and to achieve high performance of the concrete, the use of Metakaolin a relatively new approach, the chief problem with its extreme fineness and high water requirement when mixed with cement.

I. INTRODUCTION

The most common cementitious materials that are used as concrete constituents, in addition to Portland cement, are fly ash, ggbs, silica fume, and Metakaolin. Presently, the use of pozzolanic materials such as Metakaolin (MK) in Portland cement is becoming more common. They save energy, conserve resources and have many technical benefits.

Metakaolin, produced by controlled thermal treatment of kaolin, can also be used as a concrete constituents, since it has pozzolanic properties. Metakaolin is a manufactured pozzolanic mineral admixtures which significantly enhance many performance characteristics of cement based mortars, concrete and related products. The use of pozzolanic materials in the manufacturing of the concrete has a long, successful history. Most pozzolans used in the world today are by products from other industries, such as coal fly ash, blast Furnas slag, rice husk and silica fume. As such there has been relatively little work done with regard to manufactured, optimized and engineered pozzolanic materials which are specially intended for use in Portland cement based formations. The use of Metakaolin and various chemical admixtures have staple in gradients in the production of concrete with designed strength in excess of 7500psi (>50Mpa).

Among the various methods used to improve the durability of concrete against high temperature and achieve high performance of concrete the use metakaolin is relatively new approach. The partial replacement of ordinary Portland cement with pozzolanic materials can be advantageous in increasing the stability against high temperature, mortar or concrete if the proper curing regime is adopted. The due to the fact that the Calcium hydroxide produced by the cement hydration react with pozzolana and produce additional gel which has a pore blocking effect and therefore alters the pore structure and the strength. In addition there is a reduction of calcium Hydroxide (CH) which leads to improved resistance to sulphate attack and alkali-silica reactions.


PRINCIPAL

Baba Institute of Technology and Sciences
Bakkannapalem (V), Madhurawada (P)
Visakhapatnam

Study on Mechanical Properties of Concrete by using Robo Sand as a Partial Replacement of Fine Aggregate

29

Dhuda Anil

PG Student

Department of Civil Engineering

Dr.N.Victor Babu

professor, Department of Civil Engineering.

BABA Institute of Technology and Sciences, Visakhapatnam-48, India.

Abstract- Robo sand is one of the most used among such materials to replace river sand, which can be used as an alternative to fine aggregate in concrete. In the present investigation workability and strength of concrete was evaluated by replacement of natural sand by Robo sand in proportions of 0%, 50%, 75%, and 100% is studied for M25 and M35 grade concrete cubes, cylinders and prisms. Slump cone method is taken for finding workability. For strength parameters for each grade of concrete Cubes, Cylinder sand Prisms were casted and tested at the age of 7 and 28 days. In this present experimental study on concrete having grades of M25 and M35 are prepared by replacing natural sand by Robo sand. Concrete specimens were tested for evaluation of compressive strength and water absorption. The compressive strength of concrete specimens made with 50% replacement of River sand by Robo sand gives higher strength of 12% to 15% and with 100% replacement gives a higher strength of 3% to 4% as compare to reference mix. The split tensile strength of concrete specimens made with 50% replacement of river sand by Robo sand gives higher strength of 7% to 9% and with 100% replacement gives a higher strength of 3% to 4% as compare to reference mix. The flexural strength of concrete specimens made with 50% replacement of river sand by Robo sand gives higher strength of 20% to 22% and with 100% replacement gives a higher strength of 5% to 8% as compare to reference mix.

1. INTRODUCTION

With the world wide decline in the availability of construction sands along with the environmental pressure to reduce extraction of sand from rivers. In general, the demand of natural sand is quite high in developing countries to satisfy the rapid infrastructure growth, in this situation developing country like India facing shortage in good quality natural sand. Particularly in India, natural sand deposits are being depleted and causing serious threat to environment as well as society. Increasing extraction of natural sand from river beds causing many problems. Due to the effects ban on sand mining implemented by different states, and with the increasing demand for river sand for construction works, many civil engineers have expressed the need to promote use of Robo sand in the construction industry. Fine aggregate is essential component of concrete and cement mortar. So, need for clean sand in the construction from the point of view of durability of structures. As the demand for Natural River sand is surpassing the availability, has resulted in fast depletion of natural sand sources. Robo sand is the answer for this problem especially when some states have already banned the use of river sand for construction. As per reports, Robo sand is widely used all around the world and technicians of major projects around the world insist on the compulsory use of Robo sand because of its consistent gradation and zero impurity.

PRINCIPAL

Baba Institute of Technology and Sciences
Bakkannapalem (V), Madhurawada (P)
Visakhapatnam

Geological mapping for Thandava Reservoir Catchment of Visakhapatnam, Andhra Pradesh using Geo-Spatial technologies

Madhuri Mulpuri

Neela Victor Babu

B. Sridhar

Aditya Allamraju

Abstract- A geologic map provides basic data for understanding both past and present-day processes affecting a region of the Earth. This is the key to provide geologic information that can be utilized to reduce death and damage caused by geologic hazards. For the present investigation, thematic maps of geology have been generated from the IRS-ID-LISS III, dated on 11th Jan 2011. Standard visual interpretation methods given by NRSA (1996) have been adopted to portray the features on map. From this study of the Thandava reservoir, the result of geology map shows that most of the area is occupied by migmatite (67%) followed by Khondalite (20%). Further, the results of lineament map showed the main strike directions were associated with two trending NW-SE and NE-SE. This reveals that the area had been structurally controlled. These structural features have played a prominent role in relation to drainage development and erosion in the area.

Keywords: geologic map, geologic hazards, lineament map, strike directions.

1. INTRODUCTION

A geology map is a special-purpose map made to show geological features. They are depicted by color or symbols to indicate where they are exposed at the surface. All the structural features such as faults, folds, foliations, and lineaments are shown with strike and dip or trend and plunge symbols which give them the 3D orientations. The strike of the rock formations is NNE-SSW; however, local variations are observed (Sriramadas, 1964; Chetty *et al.*, 2002; Jagadeeswara Rao, 2006). A lineament is a linear feature formed at fracture zones, shear zones and igneous intrusions such as dykes that emphasizes an underlying geological structure. It may comprise of fault-aligned valley, a series of faults or fold-aligned hills, a straight coastline or indeed a combination of these features. They are often apparent in geological or topographic maps and are clearly visible on aerial photographs or as linear tonal discontinuities in satellite images. Various scientists have emphasized the importance of aerial photos and satellite imagery in structural analysis, especially in lineament mapping (Prudhvi Raju and Valdyanadhan, 1981).


PRINCIPAL
Baba Institute of Technology and Sciences
Bakkannapalem (V), Madhuravada (P)
Visakhapatnam

SYNTHESIS OF NEEM OIL ESTERS AS BIOLUBRICANT BASE STOCK

Kodanda Rama Rao Chebattina
Department of Mechanical Engineering
GITAM, Visakhapatnam, India
Email: kodandaram.nitw@gmail.com

CH Uma Chaitanya
Department of Mechanical Engineering
BABA Institute of Technology and Sciences, Visakhapatnam, India
Email: uma.nitw@gmail.com

Dharmala Venkata Padmaja
Department of Mechanical Engineering
BABA Institute of Technology and Sciences, Visakhapatnam, India
Email: padmajadv@gmail.com

Abstract- Global warming has become a worldwide concern, which is highly contributed by the emissions of IC engine combustion products. Inflation in fuel prices as well and shortage of its supply due to unsustainable energy situations worldwide owing to unequal distribution of natural wealth as well as geographical, environmental and economic status are also another concern to look for an alternative source for this. Ever increasing drift of energy consumption is due to over population, transportation and luxurious life style has motivated researches to carry out research on renewable sources as sustainable alternative. In this study renewable, carbon neutral, non-toxic, essentially free of sulphur, bio-lubricants are synthesized as alternative for fossil fuels. Trimethylol propane esters, neopentyl glycol esters, pentaerythritol esters have been developed from Neem oil as lubricant base stocks. The developed base stocks were evaluated for their physico-chemical and lubricant properties. Trimethylol propane esters were prone to be better lubricants compared to other esters synthesized.

Keywords Calophyllum oil, esters, bio-lubricant.

1. INTRODUCTION

In present days utilization of crude oil was increased due to increased needs with growing technology and population. Crude oil is non-renewable and biodegradable. Lubricant property in vegetable oils is enhanced by converting them to polyol TMP ester [1]. The hydraulic fluids based TMP esters of rapeseed oil had good cold stability, friction and wear characteristics, and resistance against oxidation at elevated temperatures [2]. The production of polyol esters by transesterification of Jatropha oil with methanol and trimethylol propane. The effects of temperature, pressure, ratio of Jatropha oil methyl ester: trimethylol propane and catalyst amount were studied in optimization experiment. [3]. Preparation of polyol esters based on rapeseed oil, olive oil and lard with neopentyl glycol (NPG) and trimethylol propane (TMP). These polyol esters were characterised with physico chemical properties [4]. Merja Lämsä, et al., studied on 2-Ethyl-1-hexanol fatty acid esters from rapeseed oil by enzymatic transesterification [5]. The performance and exhaust emission characteristics of Neem methyl ester with diesel fuel [6]. Vegetable oil is biodegradable and shows comparable performance as a replacement for mineral oil lubricants [7]. Vegetable oil-based polyol esters have been produced in various ways from FA [8,9] and FAME [10]. However, in contrast, Waara et al. [11] found that pure synthetic ester (TMP-Oleate and TMP-C8- C10) in a sliding contact resulted in high wear rates and a lot of abrasive marks on the surface. Zulkifli NWM et al [12] has studied wear prevention characteristics of a palm oil-based TMP ester as an engine lubricant [13]. The Kinetics of Transesterification of Palm-based methyl esters with Trimethylol propane. Polyol esters synthesized are eco-friendly and biodegradable which are fire resistance and have good lubricating properties [14]. Karuna et al., has studied the polyol esters of rubber seed oil as a potential source for hydraulic oils and thumba seed oil as potential bio lubricant base stocks. Polyol esters of 10-undecenoic acid and undecanoic acid are potential lubricant base stock. [15]. Tamanu based biodiesel used as non conversational fuels in compression ignition engine without any need for modification in engine design [16]. Tribiological characteristics of Neem based TMP ester as a better

SYNTHESIS OF NEEM OIL ESTERS AS BIOLUBRICANT BASE STOCK

Kodanda Rama Rao Chebattina
Department of Mechanical Engineering
GITAM Visakhapatnam, India
Email: kodandaram.nitw@gmail.com

CH Uma Chaitanya
Department of Mechanical Engineering
BABA Institute of Technology and Sciences, Visakhapatnam, India
Email: uma.nitw@gmail.com

Dharmala Venkata Padmaja
Department of Mechanical Engineering
BABA Institute of Technology and Sciences, Visakhapatnam, India
Email: padmajady@gmail.com

Abstract- Global warming has become a worldwide concern, which is highly contributed by the emissions of IC engine combustion products. Inflation in fuel prices as well and shortage of its supply due to unsustainable energy situations world-wide owing to unequal distribution of natural wealth as well as geographical, environmental and economic status are also another concern to look for an alternative source for this. Ever increasing drift of energy consumption is due to over population, transportation and luxurious life style has motivated researches to carry out research on renewable sources as sustainable alternative. In this study renewable, carbon neutral, non-toxic, essentially free of sulphur, bio-lubricants are synthesized as alternative for fossil fuels. Trimethylol propane esters, neopentyl glycol esters, pentaerythritol esters have been developed from Neem oil as lubricant base stocks. The developed base stocks were evaluated for their physico-chemical and lubricant properties. Trimethylol propane esters were found to be better lubricants compared to other esters synthesized.

Keywords – Calophyllum oil, esters, bio-lubricant.

1. INTRODUCTION

In present days utilization of crude oil was increased due to increased needs with growing technology and population. Crude oil is non-renewable and biodegradable. Lubricant property in vegetable oils is enhanced by converting them to polyol TMP ester [1]. The hydraulic fluids based TMP esters of rapeseed oil had good cold stability, friction and wear characteristics, and resistance against oxidation at elevated temperatures [2]. The production of polyol esters by transesterification of Jatropha oil with methanol and trimethylol propane. The effects of temperature, pressure, ratio of Jatropha oil methyl ester: trimethylol propane and catalyst amount were studied in optimization experiment. [3]. Preparation of polyol esters based on rapeseed oil, olive oil and lard with neopentyl glycol (NPG) and trimethylol propane (TMP). These polyol esters were characterised with physico chemical properties [4]. Merja Lämsä, et al., studied on 2-Ethyl-1-hexanol fatty acid esters from rapeseed oil by enzymatic transesterification [5]. The performance and exhaust emission characteristics of Neem methyl ester with diesel fuel [6]. Vegetable oil is biodegradable and shows comparable performance as a replacement for mineral oil lubricants [7]. Vegetable oil-based polyol esters have been produced in various ways from FA [8,9] and FAME [10]. However, in contrast, Waara et al. [11] found that pure synthetic ester (TMP-Oleate and TMP-C8- C10) in a sliding contact resulted in high wear rates and a lot of abrasive marks on the surface. Zulkifli NWM et al [12] has studied wear prevention characteristics of a palm oil-based TMP ester as an engine lubricant [13]. The Kinetics of Transesterification of Palm-based methyl esters with Trimethylol propane. Polyol esters synthesized are eco-friendly and biodegradable which are fire resistance and have good lubricating properties [14]. Karuna et al., has studied the polyol esters of rubber seed oil as a potential source for hydraulic oils and thumba seed oil as potential bio lubricant base stocks. Polyol esters of 10-undecenoic acid and undecanoic acid are potential lubricant base stock. [15]. Tamanu based biodiesel used as non conversational fuels in compression ignition engine without any need for modification in engine design [16]. Tribiological characteristics of Neem based TMP ester as a better

Analysis of Wind Turbine Blade of Composite Material Subjected to Different Loads by Identifying its Natural Frequency

Dharmala Venkata Padmaja

Department of Mechanical Engineering,

BABA Institute of Technology and Sciences, Visakhapatnam, India
padmajadv@gmail.com

Ch Uma Chaitanya

Department of Mechanical Engineering,

BABA Institute of Technology and Sciences, Visakhapatnam, India
umanitw@gmail.com

Sireesha Andiboyinaa

Department of Mechanical Engineering,

BABA Institute of Technology and Sciences, Visakhapatnam, India.
sireesha.andiboyinaa04@gmail.com

Abstract: Blades are being developed in composite materials in order to achieve low-weight and high strength construction. Composite blades are made with unidirectional fibers parallel to the blade's long axis. This project examines the modal analysis techniques applied in experiments using a uniform and a stepped beam. These simplified shapes are representative of a wind turbine blade. Natural frequencies have been identified, therefore designers can ensure those natural frequencies will not be close to the frequency of the main excitation forces in order to avoid resonance.

The turbine blade is approximated by a cantilever. Therefore, it is fully constrained where attached to a turbine shaft/hub. The specimens i.e., uniform beam is modeled in CATIA and analysis is done in ANSYS. Analysis of static, modal, harmonic and transient are carried out. The results obtained from analysis of uniform beam by changing material with steel and carbon epoxy is compared.

Keywords: Wind Turbine Blade, Natural Frequency, Composite Materials, Failure Analysis

1. INTRODUCTION

Recently, wind energy has become one of the most important forms of alternate energy sources. In a wind power production system, turbine blades are used to convert wind energy to electric energy. In general, turbine blades are required to be light in weight with high strength to withstand extreme wind loads and have a little mass moment of inertia so that they can start to rotate at low wind speed. On the other hand, a wind power system has to survive severe environment for at least 20 years. During its lifetime, a wind blade may experience strong wind with speed as high as 60m/s and in such condition, the blade should have enough strength to sustain the extreme wind load. By considering these requirements, the reliability of wind turbine blade has become an important research work. Recently, many research papers have been devoted to the study of different aspects of wind turbine blades including design, structural and theoretical analysis at different wind loads. A method established on the basis of ANSYS software on two shapes of blade and two different composite materials. Analysis is done on static, modal, harmonic and transient.

2. WIND BLADE CHARACTERISTICS

The shape of the wind turbine blade is considered as uniform beam. The wind turbine blade is modeled in CATIA and then it is imported to ANSYS with

BABA INSTITUTE OF TECHNOLOGY AND SCIENCES
Bakkannapalem (V), Madhurawada (P)
Visakhapatnam

Analysis of Wind Turbine Blade of Composite Material Subjected to Different Loads by Identifying its Natural Frequency

Dharmala Venkata Padmaja
Department of Mechanical Engineering,
BABA Institute of Technology and Sciences, Visakhapatnam, India
padmajadv@gmail.com

Ch Uma Chaitanya
Department of Mechanical Engineering,
BABA Institute of Technology and Sciences, Visakhapatnam, India
umanitw@gmail.com

Sireesha Andiboyinaa
Department of Mechanical Engineering,
BABA Institute of Technology and Sciences, Visakhapatnam, India.
sireesha.andiboyinaa04@gmail.com

Abstract: Blades are being developed in composite materials in order to achieve low-weight and high strength construction. Composite blades are made with unidirectional fibers parallel to the blade's long axis. This project examines the modal analysis techniques applied in experiments using a uniform and a stepped beam. These simplified shapes are representative of a wind turbine blade. Natural frequencies have been identified, therefore designers can ensure those natural frequencies will not be close to the frequency of the main excitation forces in order to avoid resonance.

The turbine blade is approximated by a cantilever. Therefore, it is fully constrained where attached to a turbine shaft/hub. The specimens i.e, uniform beam is modeled in CATIA and analysis is done in ANSYS. Analysis of static, modal, harmonic and transient are carried out. The results obtained from analysis of uniform beam by changing material with steel and carbon epoxy is compared.

Keywords: Wind Turbine Blade, Natural Frequency, Composite Materials, Failure Analysis

1. INTRODUCTION

Recently, wind energy has become one of the most important forms of alternate energy sources. In a wind power production system, turbine blades are used to convert wind energy to electric energy. In general, turbine blades are required to be light in weight with high strength to withstand extreme wind loads and have a little mass moment of inertia so that they can start to rotate at low wind speed. On the other hand, a wind power system has to survive severe environment for at least 20 years. During its lifetime, a wind blade may experience strong wind with speed as high as 60m/s and in such condition, the blade should have enough strength to sustain the extreme wind load. By considering these requirements, the reliability of wind turbine blade has become an important research work. Recently, many research papers have been devoted to the study of different aspects of wind turbine blades including design, structural and theoretical analysis at different wind loads. A method established on the basis of ANSYS software on two shapes of blade and two different composite materials. Analysis is done on static, modal, harmonic and transient.

2. WIND BLADE CHARACTERISTICS

The shape of the wind turbine blade is considered as uniform beam. The wind turbine blade is modeled in CATIA and then it is imported to ANSYS software with BABA Institute of Technology and Sciences
Bakkannapalem (V), Madhurawada (P)
Visakhapatnam

Analysis of Wind Turbine Blade of Composite Material Subjected to Different Loads by Identifying its Natural Frequency

Dharmala Venkata Padmaja
Department of Mechanical Engineering,
BABA Institute of Technology and Sciences, Visakhapatnam, India
padmajadv@gmail.com

Ch Uma Chaitanya
Department of Mechanical Engineering,
BABA Institute of Technology and Sciences, Visakhapatnam, India
umanitw@gmail.com

Sireesha Andiboyinaa
Department of Mechanical Engineering,
BABA Institute of Technology and Sciences, Visakhapatnam, India.
sireesha.andiboyinaa04@gmail.com

Abstract: Blades are being developed in composite materials in order to achieve low-weight and high strength construction. Composite blades are made with unidirectional fibers parallel to the blade's long axis. This project examines the modal analysis techniques applied in experiments using a uniform and a stepped beam. These simplified shapes are representative of a wind turbine blade. Natural frequencies have been identified, therefore designers can ensure those natural frequencies will not be close to the frequency of the main excitation forces in order to avoid resonance.

The turbine blade is approximated by a cantilever. Therefore, it is fully constrained where attached to a turbine shaft/hub. The specimens i.e., uniform beam is modeled in CATIA and analysis is done in ANSYS. Analysis of static, modal, harmonic and transient are carried out. The results obtained from analysis of uniform beam by changing material with steel and carbon epoxy is compared.

Keywords: Wind Turbine Blade, Natural Frequency, Composite Materials, Failure Analysis

1. INTRODUCTION

Recently, wind energy has become one of the most important forms of alternate energy sources. In a wind power production system, turbine blades are used to convert wind energy to electric energy. In general, turbine blades are required to be light in weight with high strength to withstand extreme wind loads and have a little mass moment of inertia so that they can start to rotate at low wind speed. On the other hand, a wind power system has to survive severe environment for at least 20 years. During its lifetime, a wind blade may experience strong wind with speed as high as 60m/s and in such condition, the blade should have enough strength to sustain the extreme wind load. By considering these requirements, the reliability of wind turbine blade has become an important research work. Recently, many research papers have been devoted to the study of different aspects of wind turbine blades including design, structural and theoretical analysis at different wind loads. A method established on the basis of ANSYS software on two shapes of blade and two different composite materials. Analysis is done on static, modal, harmonic and transient.

2. WIND BLADE CHARACTERISTICS

The shape of the wind turbine blade is considered as uniform beam. The wind turbine blade is modeled in CATIA and then it is imported to ANSYS Software with

PRINCIPAL
BABA Institute of Technology and Sciences
Bakkannapalem (V), Madhurawada (P)
Visakhapatnam

Conversion of Waste Plastic into Burning Oil: Review

R.Rajesh

*Assistant Professor, Department of Mechanical Engineering
Baba Institute of Technology and sciences, Visakhapatnam*

S.M.Idris

*Assistant Professor, Department of Mechanical Engineering
Baba Institute of Technology and sciences, Visakhapatnam
syedmahaboobidris@gmail.com*

Abstract: Now a day's utilization of plastics are day by day increasing. These plastics are made from crude oils through polymerization process or poly condensation process. After utilization of plastic materials those are treated as waste plastic once it enters in to the earth can't be degradable easily these are causes for very dangerous problems to human beings, all living animals in the earth. There are number of techniques to reduce this plastic pollution. In that first technique is to dump waste plastic in to the sea when this happens the living things which are in the sea will swallow so that those life will be decrease. another technique is to dump waste plastic into lands due to this the fertility of earth will decreases another technique is to burn this waste plastic with in the atmosphere this is very dangerous. because once plastic burnt this will produce very harmful gases these are causes for health problems for humans.

To avoid these problems one technique is generally used for to convert this waste plastic in to burning fuel that technique is pyrolysis process. In this process waste plastic is generally heated in the absence of oxygen and gets burning oil. The main objective of this paper is to explain about pyrolysis technique.

Keywords: waste plastic, pyrolysis process, burning oil.

1. INTRODUCTION

Plastics play an vital part in day- nowadays life. It is interesting fabric since of their durability, light weight, resistance to water and chemicals, safe to warm and cold, moo electrical and warm conductivity, ease of creation, exceptional color extend, more plan adaptability, strength and vitality effectiveness. Due to over properties it is utilized in bundling materials, agribusiness, development, separator, car segment, electronic gadgets, materials and sports hardware and toys. Plastics constitutes in two primary categories. It is thermoplastics and thermoset plastics. Thermoplastics make up 80% of the plastics and thermoset plastics make up of remaining 20% of plastics delivered nowadays (Birley et al, 1988), etc. Thermo plastics can re-melt or remold and so it recyclable effectively but thermoset plastics cannot re-melt or reshape and so it is troublesome to reusing. Utilize of diverse sort of a few thermo plastics is given in table underneath. Plastics are generally cheap, simple accessible, simple to fabricate and their flexibility supplant to customary materials. Plastic squander


PRINCIPAL
Baba Institute of Technology and Sciences
Bakkannapalem (V), Madhurawada (P)
Visakhapatnam

Conversion of Waste Plastic into Burning Oil: Review

R.Rajesh

*Assistant Professor, Department of Mechanical Engineering
Baba Institute of Technology and sciences, Visakhapatnam*

S.M.Idris

*Assistant Professor, Department of Mechanical Engineering
Baba Institute of Technology and sciences, Visakhapatnam
syedmahaboobidris@gmail.com*

Abstract: Now a day's utilization of plastics are day by day increasing. These plastics are made from crude oils through polymerization process or poly condensation process. After utilization of plastic materials those are treated as waste plastic once it enters in to the earth can't be degradable easily these are causes for very dangerous problems to human beings, all living animals in the earth. There are number of techniques to reduce this plastic pollution. In that first technique is to dump waste plastic in to the sea when this happens the living things which are in the sea will swallow so that those life will be decrease. another technique is to dump waste plastic into lands due to this the fertility of earth will decreases another technique is to burn this waste plastic with in the atmosphere this is very dangerous. because once plastic burnt this will produce very harmful gases these are causes for health problems for humans.

To avoid these problems one technique is generally used for to convert this waste plastic in to burning fuel that technique is pyrolysis process. In this process waste plastic is generally heated in the absence of oxygen and gets burning oil. The main objective of this paper is to explain about pyrolysis technique.

Keywords: waste plastic, pyrolysis process, burning oil.

1. INTRODUCTION

Plastics play an vital part in day- nowadays life. It is interesting fabric since of their durability, light weight, resistance to water and chemicals, safe to warm and cold, moo electrical and warm conductivity, ease of creation, exceptional color extend, more plan adaptability, strength and vitality effectiveness. Dueto over propertiesitis utilized in bundling materials, agribusiness, development, separator, car segment, electronic gadgets, materials and sports hardware and toys. Plastics constitutes in two primary categories. It is thermoplastics and thermoset plastics. Thermoplastics make up 80% of the plastics and thermoset plastics make up of remaining 20% of plastics delivered nowadays (Birley et al, 1988), etc. Thermo plastics can re-melt or remold and so it recyclable effectively but thermoset plastics cannot re-melt or reshape and so it is troublesome to reusing. Utilize of diverse sort of a few thermo plastics is given in table1 underneath. Plastics are generally cheap, simple accessible, simple to fabricate andtheir flexibility supplant to customary materials. Plastic squander



PRINCIPAL
Baba Institute of Technology and Sciences
Bakkannapalem (V), Madhurawada (P)
Visakhapatnam

Operational and Productivity Performance of Andhra Pradesh State Financial Corporation: An Experimental Analysis

Dr. P. Sanjeevi,

*Professor, Department of Management Studies,
BITS VSP*

Mr. M. Vasudeva Rao,

*Assistant Professor, DMS,
BITS VSP*

MR. S. Seeman Dutta,

*Assistant Professor, DMS,
BITS VSP*

Abstract

Indian banking industry has transformed in recent years due to globalization in the world market, which has resulted in fierce competition. In this paper, an attempt has been made to analyze the operational performance and productivity performance of Andhra Pradesh State Financial Corporation. Analytical frame of the operational aspects in terms of the sanctions, disbursements, collection, operating profit, number of accounts and number of employees and in general, and to different sectors, purposes; and also, the trends over the five years period under study have been presented. In addition, the operational and productivity performance is evaluated by the known parameters and bench marks. The study is based on secondary data collected from annual reports of APSEC for the study period i.e. 2013-14 to 2017-18. Accounting Ratios, t-test, correlation analysis has been used to analyze the present data. The study suggested that APSEC should highlight on generating more profits by efficient utilization of its assets, capital and improving the productive efficiency of number of accounts and employees.

Key words: Productivity Performance, Operational Performance, Sanctions, Disbursement, collection, operating profit.

INTRODUCTION

The mandate given in the statute drives the corporation to develop entrepreneurship, bring in balanced regional development and drive the industrialization in the state in order to bring the economic

CONSUMER ATTITUDE TOWARDS THE PRODUCTS OF HYPER MARKETS IN VISAKHAPATNAM

Dr. B. Rama Jyothi*

Mr. M. Vasudeva Rao**

Mr. Naresh Kumar Kuppli***

Abstract

Market is a place where buyers and sellers meet, goods and services are offered for sale and transfer of ownership occurs. Sometimes a market is a precisely physical place and sometimes it refers to groups of people. At present day's hyper markets plays an important role to develop the modern markets and improve the customer services and promote the economic excelleration. As market consists of consumers, their attitude, beliefs, loyalty and perception is important in modern marketing. Without customers, a firm has no revenues, no profits, and therefore no market value. Consumers all have different tastes, like and dislikes and adopt different behaviour patterns while making purchase decisions. The present study deals with the attitude, consumer satisfaction and preferences with regard to the services, prices, quality, products etc. of hyper markets.

Keywords:

Modern Markets;
Customer Attitude;
Customer satisfaction;
Hyper markets;

Copyright © 201x International Journals of Multidisciplinary
Research Academy. All rights reserved.

Author correspondence:

Dr. B. Rama Jyothi,
Assistant Professor,
Baba Institute of Technology and Sciences,
PM palem,
Visakhaptnam-48.
Email: bramajyothi@bitsvizag.com

Mr. M. Vasudeva Rao**
Assistant Professor,
Baba Institute of Technology and Sciences,
PM palem,
Visakhaptnam-48.
Email: vasumalicherla@gmail.com

Mr. Naresh Kumar Kuppli***
Assistant Professor,
Baba Institute of Technology and Sciences,
PM palem,
Visakhaptnam-48.
Email: nareshpatnaik.k@gmail.com

* Doctorate Program, Linguistics Program Studies, Udayana University Denpasar, Bali-Indonesia (9 pt)
** STIMIK STIKOM-Bali, Renon, Denpasar, Bali-Indonesia



International Journal of Engineering, Science and Mathematics

Vol. x issue x, Month 201x,

ISSN: 2320-0294 Impact Factor: 6.765

Journal Homepage: <http://www.ijesmj.co.in>, Email: ijesmj@gmail.com

Double-Blind

Peer Reviewed Refereed Open Access International Journal - Included in the International Serial Directories Indexed & Listed at Ulrich's Periodicals Directory ©, U.S.A., Open J-Gate as well as in Cabell's Directories of Publishing Opportunities, U.S.A.

CONSUMER ATTITUDE TOWARDS THE PRODUCTS OF HYPER MARKETS IN VISAKHAPATNAM

Dr. B. Rama Jyothi*

Mr. M. Vasudeva Rao**

Mr. Naresh Kumar Kuppli***

Abstract

Market is a place where buyers and sellers meet, goods and services are offered for sale and transfer of ownership occurs. Sometimes a market is a precisely physical place and sometimes it refers to groups of people. At present day's hyper markets plays an important role to develop the modern markets and improve the customer services and promote the economic excelleration. As market consists of consumers, their attitude, beliefs, loyalty and perception is important in modern marketing. Without customers, a firm has no revenues, no profits, and therefore no market value. Consumers all have different tastes, like and dislikes and adopt different behaviour patterns while making purchase decisions. The present study deals with the attitude, consumer satisfaction and preferences with regard to the services, prices, quality, products etc. of hyper markets.

Keywords:

Modern Markets;
Customer Attitude;
Customer satisfaction;
Hyper markets;

Copyright © 201x International Journals of Multidisciplinary Research Academy. All rights reserved.

Author correspondence:

Dr. B. Rama Jyothi,
Assistant Professor,
Baba Institute of Technology and Sciences,
PM palem,
Visakhapatnam-48.
Email: bramajyothi@bitsvizag.com

Mr. M. Vasudeva Rao**
Assistant Professor,
Baba Institute of Technology and Sciences,
PM palem,
Visakhapatnam-48.
Email: vasumalicherla@gmail.com

Mr. Naresh Kumar Kuppli***
Assistant Professor,
Baba Institute of Technology and Sciences,
PM palem,
Visakhapatnam-48.
Email: nareshpatnaik.k@gmail.com

* Doctorate Program, Linguistics Program Studies, Udayana University Denpasar, Bali-Indonesia (9 pt)

** STIMIK STIKOM-Bali, Renon, Depasar, Bali-Indonesia

RURAL WOMEN ENTREPRENEURS ROLE &CHANGE IN THE SOCIETY: CASE STUDY

Dr. SREENIVASA RAO BEHARA

Professor, Department of Management Studies,
BABA Institute of Technology and Sciences (BITS),
PM Palem, Visakhapatnam- 500 048

MR P.B. NARENDRA KIRAN

Head, Department of Management Studies,
BABA Institute of Technology and Sciences (BITS),
PM Palem, Visakhapatnam- 500 048

Abstract:

A society that does not have an optimistic, positive empowering image of the future is endangered. When society is skewed & power, it begins an entropic spiral to isolation – Entrepreneurs can their skills to restructure the society. Society entrepreneurs are change agents that improve society by new models & developing equal effective often less hierarchical complex than existing ones. Society entrepreneurship mainly defined by Researchers and practitioners in various ways with common factor that adds value to a community mission through innovative, risk-taking, similar business practices. Society entrepreneurship combines innovation with the importance of community needs and their work overlaps with social justice and environmental presentation movements. Ancient Indian Women were House Wives and were accorded to follow their husbands in their religious duties. Gradually the Change came into varying social-political balance, women were changed their background aggressively dramatic exploitation.

Keywords: empowerment, entrepreneur & innovation.

RURAL WOMEN ENTREPRENEURS ROLE & CHANGE IN THE SOCIETY: CASE STUDY

Dr. SREENIVASA RAO BEHARA

Professor, Department of Management Studies,
BABA Institute of Technology and Sciences (BITS),
PM Palem, Visakhapatnam- 500 048

MR P.B. NARENDRA KIRAN

Head, Department of Management Studies,
BABA Institute of Technology and Sciences (BITS),
PM Palem, Visakhapatnam- 500 048

Abstract:

A society that does not have an optimistic, positive empowering image of the future is endangered. When society is skewed & power, it begins an entropic spiral to isolation – Entrepreneurs can their skills to restructure the society. Society entrepreneurs are change agents that improve society by new models & developing equal effective often less hierarchical complex than existing ones. Society entrepreneurship mainly defined by Researchers and practitioners in various ways with common factor that adds value to a community mission through innovative, risk-taking, similar business practices. Society entrepreneurship combines innovation with the importance of community needs and their work overlaps with social justice and environmental presentation movements. Ancient Indian Women were House Wives and were accorded to follow their husbands in their religious duties. Gradually the Change came into varying social-political balance, women were changed their background aggressively dramatic exploitation.

Keywords: empowerment, entrepreneur & innovation.


PRINCIPAL
Baba Institute of Technology and Sciences
Bakkannapalem (V), Madhurawada (P)
Visakhapatnam



TRAINING AND DEVELOPMENT PRACTICES IN SELECT CORPORATE RETAIL STORES IN VISAKHAPATNAM

Mr. P.BalaNarendraKiran

Head, Department of Management Studies
Baba Institute of Technology and Sciences,
Visakhapatnam, Andhrapradesh

K.Sireesha

Assistant Professor,
Department of Management Studies
Baba Institute of Technology and Sciences,
Visakhapatnam, Andhrapradesh

Abstract:

A profound training program acts as a vehicle to enhance employee skills and enable them to perform better in their job. Training and development is very crucial to the employees, the organization and their effectiveness. Training is important in retailing because majority of the employees have direct contact with customers. They are responsible for helping customers, satisfy their needs and resolve their problems. Training is important not only from the point of view of employees, but also for the organization. This study has examined the perception of respondents on the present training and development practices of select corporate retail organizations. With this background, this study is conducted to know the perception of respondents in terms of their demographic profiles of gender, age, education and social status.

Keywords: Training, Development, Promotion

1. Introduction

Training programmes are created not only for the new recruits but also often designed for the existing personnel within the organization especially when they are either transferred or promoted. All large and small retail organizations must conduct some form of a training programme to be able to perform smoothly. This is a part of the overall induction programme that is exclusively designed for every position. A comprehensive training and development program helps "in deliberating on the knowledge, skills and attitudes necessary

TRAINING AND DEVELOPMENT PRACTICES IN SELECT CORPORATE RETAIL STORES IN VISAKHAPATNAM

Mr. P.BalaNarendraKiran

Head, Department of Management Studies
Baba Institute of Technology and Sciences,
Visakhapatnam, Andhrapradesh

K.Sireesha

Assistant Professor,
Department of Management Studies
Baba Institute of Technology and Sciences,
Visakhapatnam, Andhrapradesh

Abstract:

A profound training program acts as a vehicle to enhance employee skills and enable them to perform better in their job. Training and development is very crucial to the employees, the organization and their effectiveness. Training is important in retailing because majority of the employees have direct contact with customers. They are responsible for helping customers, satisfy their needs and resolve their problems. Training is important not only from the point of view of employees, but also for the organization. This study has examined the perception of respondents on the present training and development practices of select corporate retail organizations. With this background, this study is conducted to know the perception of respondents in terms of their demographic profiles of gender, age, education and social status.

Keywords: Training, Development, Promotion

1. Introduction

Training programmes are created not only for the new recruits but also often designed for the existing personnel within the organization especially when they are either transferred or promoted. All large and small retail organizations must conduct some form of a training programme to be able to perform smoothly. This is a part of the overall induction programme that is exclusively designed for every position. A comprehensive training and development program helps “in deliberating on the knowledge, skills and attitudes necessary

A Geo-Spatial approach to the Land Use/Land Cover mapping of the centre part of Srikakulam district, Andhra Pradesh, India

B. Sridhar

Department of Geo-Engineering

Andhra University College of Engineering (A) Visakhapatnam, Andhra Pradesh, India
sridhar.bendalam@gmail.com, Tel.: +91-8341556390

P Jagadeeswara Rao

Department of Geo-Engineering

Andhra University College of Engineering (A) Visakhapatnam, Andhra Pradesh, India
Neela Victor Babu

Baba Institute of Technology and Science, Visakhapatnam

Aditya Allamraju

Department of Geo-Engineering

Andhra University College of Engineering (A) Visakhapatnam, Andhra Pradesh, India

Abstract-In the present investigation, the land use/land cover mapping of the centre part of Srikakulam district in the state of Andhra Pradesh has been carried out using Resourcesat-1, LISS IV satellite imagery based on standard visual interpretation techniques. The identified land use / land cover features are Aquaculture, Coastal sand, Coastal wetland, Mud-flat, Coastal-wetland, Creek, Deciduous dense forest, Deciduous open forest, Forest-plantations, Industrial, Kharif (single crop), Kharif +Rabi(double-cropped), Plantation, Tank, Towns/cities(Urban), Villages(Rural), Wastelands, Barren-rock, Wastelands, Gullied land, Wastelands, hills with dense vegetation, Wastelands-hills with scrub, Land without-scrub, Salt affected land and Stone-waste. Out of all these features, the maximum area is covered by Kharif +Rabi (double-cropped) (28.091%) followed by Kharif (single crop) (22.773%), Plantation (13.13 %) and Wastelands, hills with scrub (10.916%).

Keywords: Land Use, Land Cover, Resourcesat-1, LISS IV, satellite imagery, Visual Interpretation

1. INTRODUCTION

Land and water have been the basic elements of life supporting systems on our planet since the dawn of civilization (Redclift, M. 2002). All great civilizations flourished where these resources were available in plenty and they declined or perished with the depletion of these resources. In recent years, the land resource has been subjected to a variety of pressures, yet it is surviving and sustaining mankind. The concerned matter is the way, land is being overexploited which is leading to threat to the environment. Out of all the species on the earth, man is the chief culprit to this degradation. He views land in terms of its utility to meet his perceived needs and wants. The most easily categorized varieties of land from the utility point of view are land fit for use, land with potential for use and land which appear useless in the foreseeable future.

Here probably lies the genesis of the problem of land degradation and erosion of ecosystems. Mahatma Gandhi said - "The Earth has enough for everybody's need but not for anybody's greed". Preserving, protecting and defending the land resources have been part of our age-old culture. The respect for the importance of land resources is best depicted in the conventional concept of Panchabhutas - land, water, fire, sky and air that

Baba Institute of Technology and Sciences
Bakkannapalem (V), Madhurawada (P)
Visakhapatnam

Geo-spatial analysis of HIV Status in East Godavari District, Andhra Pradesh, India

Dr.K Anand RatnaKumar
Dept.of Geography
Andhra University, Visakhapatnam - 530 003

Suribabu Boyidi
Senior Research Fellow
Dept of Geo Engg AUCE(A)- Andhra University

Dr.N Victor Babu
Professor &HOD,
Dept of Civil, BITS Vishakapatnam, .

Abstract: East Godavari district is one of the hardest hit states of HIV of Andhra Pradesh State in Southern India. Most part of the district is infested by HIV incidence. The year-wise data on HIV disease collected from the records of District Leprosy Office, Kakinada for the year 2002-2008 and cumulative childrens household's HIV cases for fourteen mandals of the district were collected from St.Paul's Trust, Samalkot from July, 2007 to March, 2009. Analysis of the data indicated that a total of 8561 cases were identified in the district. The analysis of HIV disease reveals that the number of HIV positive cases are steadily increasing since 2002-2003. The number was doubled during 2004-2005 and became more than 8000 by 2007-2008. Data is not available most part of the district and are available for fourteen mandals namely Samalkot, Biccavolu, Anaparthi, Mandapeta, Peddapuram, Rangampeta, Rajanagaram, Jaggampeta, Candepalle, Kakinada (Rural), Kakinada (Urban), Karapa, Pedapudi and Rayavaram. Among the mandals Peddapuram, Samalkot, Jaggampeta, Biccavolu and Mandapeta are highly prevalence of the disease low incidence in the mandals namely Rangampeta, Rayavaram and Karapa. Peddapuram records the high incidence of the disease with 1411 cases and followed by Samalkot and Jaggampeta. The maximum deaths were recorded in these mandals namely Peddapuram, Biccavolu and Rayavaram. Peddapuram records the high mortality with 35 and followed by Biccavolu and Rayavaram and less number of deaths recorded at Rangampeta. In the present study the spatial patterns of HIV incidence and mortality were made.

Introduction

Health is a state of complete, physical, mental and social well-being and not merely an absence of disease (WHO report 1948). The term disease is associated with health hazard that literally means 'without ease' (dis-ease). Park and Park (1977) defined disease as a condition in which body health is impaired, an alteration of the human body interrupting the performance of vital functions.

Acquired Immunodeficiency Syndrome (AIDS) represents the late clinical stage of infection with the Human Immunodeficiency Retro Viruses (HIV). The virus causing AIDS was first identified and called as 'lymphadenopathy-associated virus' (LAV) by a French scientists. United States researchers named it as 'human T-cell lymphotropic virus -III (HTLV-III). In May 1986, the International Committee on Taxonomy gave it new name as human immunodeficiency virus (HIV) (Anupa Mukhopadhyay et al., 1996). It has emerged as leading morbidity and premature mortality in cities around the world in a short period of over a decade. In North America it had become the principal killer of young adult males and it has outstripped overdose, suicide, homicide and hepatitis B (Anupa Mukhopadhyay et al, 2004) AIDS most often results in progressive damage to the immune, organ and central nervous systems.

Acquired Immuno Deficiency Syndrome A - Acquired (something that you get) I - Immune (capability of the human body to fight diseases)

PRINCIPAL
Baba Institute of Technology and Sciences
Bakkannapalem (V), Madhurawada (P)
Visakhapatnam

Page No:5919

A STUDY ON IDENTIFICATION OF LANDSLIDE ZONES ALONG THE RECENT EXPANSION OF GVMC, VISAKHAPATNAM DISTRICT-A GIS APPROACH

Neela Victor Babu

*Associative Professor, Department of Civil Engineering
Baba Institute of Technology and Science, Visakhapatnam
neelavictorbabu@gmail.com*

Ramojirao Jogi

*Department of Geo-Engineering, College of Engineering (A),
Andhra University, Visakhapatnam*

Gudikandhula Narasimha Rao

*Department of Geo-Engineering, College of Engineering (A),
Andhra University, Visakhapatnam*

Venkateswarlu Sunkari

*Assistant Professor, Center of ITSC, Addis Ababa Institute of technology,
Addis Ababa University, Addis Ababa, Ethiopia*

Abstract: The Greater Visakhapatnam Municipal Corporation (GVMC) city forms the area of the study in the present investigation. The population in the city is increasing at a rapid pace due to migration from surrounding areas, with corresponding increase in sub-urban areas also. The extensive growth of residential apartments along the coastal stretch resulted in excess withdrawal of groundwater, initiating seawater intrusion in these tracts. All these factors combined, leading to an acute environmental crisis in the area. Forecasting urban water demand can be of use in the management of water utilities (Salvatore Campisi-Pinto, 2012). Hence, it is necessary to manage the available water resources in an efficient manner. This is done using advanced technologies like Remote Sensing and Geographical Information System. The sites for water harvesting structures and techniques like check dams, percolation ponds, infiltration wells, afforestation, contour trenching etc.. previous literature gives us the techniques of Image Interpretation, concepts of remote sensing which help in identifying different features in the study area. Papers related to Geographic Information System tell us about the concepts of GIS which help in overlaying different layers for selecting the harvesting structures.

Keywords: LANDSLIDE ZONES, GVMC, R.S and GIS, INVESTIGATION

I. INTRODUCTION

Water being one of the most important natural resources need to be conserved and managed properly. As water covers 71 percent of the Earth's surface only three percent is fresh water and the rest 97 percent of the water on the Earth is salt water. 0.687 percent of total water resources are available as

PRINCIPAL

**Baba Institute of Technology and Sciences
Bakkannapalem (V), Madhurawada (P)
Visakhapatnam**

Application of Geospatial Technologies in Geology mapping of Bheemili in the district of Visakhapatnam, Andhra Pradesh, India

B. Sridhar

*Department of Geo-Engineering, Andhra University College of Engineering (A), Visakhapatnam
sridhar.bendalam@gmail.com, Tel.: +91-8341556390*

Neela Victor Babu

Baba Institute of Technology and Science, Visakhapatnam

Aditya Allamraju

Abstract- A geological map gives primary data for analyzing both past and present day to day processes affecting a region on the Earth. This kind of information is vital for providing information relating to geology that can aid to minimize damage and death caused by geologic hazards such as earthquakes and landslides. In the present investigation, thematic maps of geology have been produced from the IRS-ID-LISS III, 2004 and standard visual interpretation methods have been followed to portray on-screen digitations of the features. This study shows that three major kinds of rocks are exposed in the area, out of which khondalite is in majority followed by Quartzite and then Charnockite. The structural trend of rocks is NNE-SSW considering the local variations at few places. These features play an important role with respect to drainage development and erosion in the area.

Keywords: Geological hazards, Khondalite, Quartzite, Charnockite, Erosion

1. INTRODUCTION

GIS is a new and effective tool in geological mapping that helps the surveyors to generate 3-D maps of any area with accurate and desired scaling. In the present analysis, the geological thematic layers have been studied to identify the geological vulnerable areas along the road management. Standard on-screen interpretation techniques as per the nomenclature proposed by NRSA, have been followed and delineated the features on-screen. These values are obtained to understand the rock and soil behavior in sub-aerial erosions. The road constructed along the sea shore, thus the influence of marine activity on the road is also explored to locate areas vulnerable under marine conditions. The rocks along the coast are observed as khondalite, leptynite that belong to the metamorphic group. In general, the trend coincides with the trend of Eastern Ghats. The khondalite appears red to brown color rich in sillimanite, garnets and occasional occurrence of graphite, biotitic, etc. in subordinate amount. Huge red kanker or talus material has been observed along the road cutting. This material is liable to slide on to the road. Bad land topography is famous in the area. Highly gullied land masses with sparse vegetation, red soils, and quick sands are the hallmarks of the area. A number of non-perennial streams erode the area into bad land topography.


PRINCIPAL
Baba Institute of Technology and Sciences
Bakkannapalem (V), Madhuravada (P)
Visakhapatnam

STUDY ON M25 AND M30 GRADE CONCRETE BY PARTIAL REPLACEMENT OF CEMENT WITH METAKAOLIN

Ch.Kameswara Rao

Asst.Prof, Pydah College of Engineering & Technology

Dr.N.Victor Babu

Professor & H.O.D., Dept. of Civil Engg.,

BABA Institute of Technology and Sciences-Visakhapatnam

M.Sumesh Raja

Asst.Prof, Pydah College of Engineering & Technology

Abstract- Now a days the partially replacement of Ordinary Portland cement with pozzolana has been increasing widely. Most commonly used pozzolana are fly ash, silica fume, metakaolin, and ground granulated blast furnace slag. A pozzolana is a material which, when combined with calcium, exhibits cementitious properties. Pozzolana are commonly used as an addition to cement concrete mixture to increase the long-term strength and other material properties of cement concrete and in some cases to reduce the material cost of concrete. Recent researches aimed at energy conservation in the cement and concrete industry has in part, focused on the use of less energy intensive materials such as fly ash, slag and silica fume. Of the some attention has been given to the use of natural pozzolana like Metakaolin as possible partial replacement for cement. Amongst the various methods used to improve the strength, Durability and resisting power of the temperature, and to achieve high performance of the concrete, the use of Metakaolin a relatively new approach, the chief problem with its extreme fineness and high water requirement when mixed with cement.

I. INTRODUCTION

The most common cementitious materials that are used as concrete constituents, in addition to Portland cement, are fly ash, ggbs, silica fume, and Metakaolin. Presently, the use of pozzolanic materials such as Metakaolin (MK) in Portland cement is becoming more common. They save energy, conserve resources and have many technical benefits.

Metakaolin, produced by controlled thermal treatment of kaolin, can also be used as a concrete constituents, since it has pozzolanic properties. Metakaolin is a manufactured pozzolanic mineral admixtures which significantly enhance many performance characteristics of cement based mortars, concrete and related products. The use of pozzolanic materials in the manufacturing of the concrete has a long, successful history. Most pozzolans used in the world today are by products from other industries, such as coal fly ash, blast Furnas slag, rice husk and silica fume. As such there has been relatively little work done with regard to manufactured, optimized and engineered pozzolanic materials which are specially intended for use in Portland cement based formations. The use of Metakaolin and various chemical admixtures have staple in gradients in the production of concrete with designed strength in excess of 7500psi (>50Mpa).

Among the various methods used to improve the durability of concrete against high temperature and achieve high performance of concrete the use metakaolin is relatively new approach. The partial replacement of ordinary Portland cement with pozzolanic materials can be advantageous in increasing the stability against high temperature, mortar or concrete if the proper curing regime is adopted. The due to the fact that the Calcium hydroxide produced by the cement hydration react with pozzolana and produce additional gel which has a pore blocking effect and therefore alters the pore structure and the strength. In addition there is a reduction of calcium Hydroxide (CH) which leads to improved resistance to sulphate attack and alkali-silica reactions.


PRINCIPAL

Baba Institute of Technology and Sciences
Bakkannapalem (V), Madhurawada (P)
Visakhapatnam

Study on Mechanical Properties of Concrete by using Robo Sand as a Partial Replacement of Fine Aggregate

29

Dhuda Anil

PG Student

Department of Civil Engineering

Dr.N.Victor Babu

professor, Department of Civil Engineering.

BABA Institute of Technology and Sciences, Visakhapatnam-48, India.

Abstract- Robo sand is one of the most used among such materials to replace river sand, which can be used as an alternative to fine aggregate in concrete. In the present investigation workability and strength of concrete was evaluated by replacement of natural sand by Robo sand in proportions of 0%, 50%, 75%, and 100% is studied for M25 and M35 grade concrete cubes, cylinders and prisms. Slump cone method is taken for finding workability. For strength parameters for each grade of concrete Cubes, Cylinder sand Prisms were casted and tested at the age of 7 and 28 days. In this present experimental study on concrete having grades of M25 and M35 are prepared by replacing natural sand by Robo sand. Concrete specimens were tested for evaluation of compressive strength and water absorption. The compressive strength of concrete specimens made with 50% replacement of River sand by Robo sand gives higher strength of 12% to 15% and with 100% replacement gives a higher strength of 3% to 4% as compare to reference mix. The split tensile strength of concrete specimens made with 50% replacement of river sand by Robo sand gives higher strength of 7% to 9% and with 100% replacement gives a higher strength of 3% to 4% as compare to reference mix. The flexural strength of concrete specimens made with 50% replacement of river sand by Robo sand gives higher strength of 20% to 22% and with 100% replacement gives a higher strength of 5% to 8% as compare to reference mix.

1. INTRODUCTION

With the world wide decline in the availability of construction sands along with the environmental pressure to reduce extraction of sand from rivers. In general, the demand of natural sand is quite high in developing countries to satisfy the rapid infrastructure growth, in this situation developing country like India facing shortage in good quality natural sand. Particularly in India, natural sand deposits are being depleted and causing serious threat to environment as well as society. Increasing extraction of natural sand from river beds causing many problems. Due to the effects ban on sand mining implemented by different states, and with the increasing demand for river sand for construction works, many civil engineers have expressed the need to promote use of Robo sand in the construction industry. Fine aggregate is essential component of concrete and cement mortar. So, need for clean sand in the construction from the point of view of durability of structures. As the demand for Natural River sand is surpassing the availability, has resulted in fast depletion of natural sand sources. Robo sand is the answer for this problem especially when some states have already banned the use of river sand for construction. As per reports, Robo sand is widely used all around the world and technicians of major projects around the world insist on the compulsory use of Robo sand because of its consistent gradation and zero impurity.

PRINCIPAL

Baba Institute of Technology and Sciences
Bakkannapalem (V), Madhurawada (P)
Visakhapatnam

Geological mapping for Thandava Reservoir Catchment of Visakhapatnam, Andhra Pradesh using Geo-Spatial technologies

Madhuri Mulpuri

Neela Victor Babu

B. Sridhar

Aditya Allamraju

Abstract- A geologic map provides basic data for understanding both past and present-day processes affecting a region of the Earth. This is the key to provide geologic information that can be utilized to reduce death and damage caused by geologic hazards. For the present investigation, thematic maps of geology have been generated from the IRS-ID-LISS III, dated on 11th Jan 2011. Standard visual interpretation methods given by NRSA (1996) have been adopted to portray the features on map. From this study of the Thandava reservoir, the result of geology map shows that most of the area is occupied by migmatite (67%) followed by Khondalite (20%). Further, the results of lineament map showed the main strike directions were associated with two trending NW-SE and NE-SE. This reveals that the area had been structurally controlled. These structural features have played a prominent role in relation to drainage development and erosion in the area.

Keywords: geologic map, geologic hazards, lineament map, strike directions.

1. INTRODUCTION

A geology map is a special-purpose map made to show geological features. They are depicted by color or symbols to indicate where they are exposed at the surface. All the structural features such as faults, folds, foliations, and lineaments are shown with strike and dip or trend and plunge symbols which give them the 3D orientations. The strike of the rock formations is NNE-SSW; however, local variations are observed (Sriramadas, 1964; Chetty *et al.*, 2002; Jagadeeswara Rao, 2006). A lineament is a linear feature formed at fracture zones, shear zones and igneous intrusions such as dykes that emphasizes an underlying geological structure. It may comprise of fault-aligned valley, a series of faults or fold-aligned hills, a straight coastline or indeed a combination of these features. They are often apparent in geological or topographic maps and are clearly visible on aerial photographs or as linear tonal discontinuities in satellite images. Various scientists have emphasized the importance of aerial photos and satellite imagery in structural analysis, especially in lineament mapping (Prudhvi Raju and Valdyanadhan, 1981).


PRINCIPAL
Baba Institute of Technology and Sciences
Bakkannapalem (V), Madhuravada (P)
Visakhapatnam

SYNTHESIS OF NEEM OIL ESTERS AS BIOLUBRICANT BASE STOCK

Kodanda Rama Rao Chebattina
Department of Mechanical Engineering
GITAM, Visakhapatnam, India
Email: kodandaram.nitw@gmail.com

CH Uma Chaitanya
Department of Mechanical Engineering
BABA Institute of Technology and Sciences, Visakhapatnam, India
Email: uma.nitw@gmail.com

Dharmala Venkata Padmaja
Department of Mechanical Engineering
BABA Institute of Technology and Sciences, Visakhapatnam, India
Email: padmajadv@gmail.com

Abstract- Global warming has become a worldwide concern, which is highly contributed by the emissions of IC engine combustion products. Inflation in fuel prices as well and shortage of its supply due to unsustainable energy situations worldwide owing to unequal distribution of natural wealth as well as geographical, environmental and economic status are also another concern to look for an alternative source for this. Ever increasing drift of energy consumption is due to over population, transportation and luxurious life style has motivated researches to carry out research on renewable sources as sustainable alternative. In this study renewable, carbon neutral, non-toxic, essentially free of sulphur, bio-lubricants are synthesized as alternative for fossil fuels. Trimethylol propane esters, neopentyl glycol esters, pentaerythritol esters have been developed from Neem oil as lubricant base stocks. The developed base stocks were evaluated for their physico-chemical and lubricant properties. Trimethylol propane esters were found to be better lubricants compared to other esters synthesized.

Keywords Calophyllum oil, esters, bio-lubricant.

1. INTRODUCTION

In present days utilization of crude oil was increased due to increased needs with growing technology and population. Crude oil is non-renewable and biodegradable. Lubricant property in vegetable oils is enhanced by converting them to polyol TMP ester [1]. The hydraulic fluids based TMP esters of rapeseed oil had good cold stability, friction and wear characteristics, and resistance against oxidation at elevated temperatures [2]. The production of polyol esters by transesterification of Jatropha oil with methanol and trimethylol propane. The effects of temperature, pressure, ratio of Jatropha oil methyl ester: trimethylol propane and catalyst amount were studied in optimization experiment. [3]. Preparation of polyol esters based on rapeseed oil, olive oil and lard with neopentyl glycol (NPG) and trimethylol propane (TMP). These polyol esters were characterised with physico chemical properties [4]. Merja Lämsä, et al., studied on 2-Ethyl-1-hexanol fatty acid esters from rapeseed oil by enzymatic transesterification [5]. The performance and exhaust emission characteristics of Neem methyl ester with diesel fuel [6]. Vegetable oil is biodegradable and shows comparable performance as a replacement for mineral oil lubricants [7]. Vegetable oil-based polyol esters have been produced in various ways from FA [8,9] and FAME [10]. However, in contrast, Waara et al. [11] found that pure synthetic ester (TMP-Oleate and TMP-C8- C10) in a sliding contact resulted in high wear rates and a lot of abrasive marks on the surface. Zulkifli NWM et al [12] has studied wear prevention characteristics of a palm oil-based TMP ester as an engine lubricant [13]. The Kinetics of Transesterification of Palm-based methyl esters with Trimethylol propane. Polyol esters synthesized are eco-friendly and biodegradable which are fire resistance and have good lubricating properties [14]. Karuna et al., has studied the polyol esters of rubber seed oil as a potential source for hydraulic oils and thumba seed oil as potential bio lubricant base stocks. Polyol esters of 10-undecenoic acid and undecanoic acid are potential lubricant base stock. [15]. Tamanu based biodiesel used as non conversational fuels in compression ignition engine without any need for modification in engine design [16]. Tribiological characteristics of Neem based TMP ester as a better

SYNTHESIS OF NEEM OIL ESTERS AS BIOLUBRICANT BASE STOCK

Kodanda Rama Rao Chebattina
Department of Mechanical Engineering
GITAM Visakhapatnam, India
Email: kodandaram.nitw@gmail.com

CH Uma Chaitanya
Department of Mechanical Engineering
BABA Institute of Technology and Sciences, Visakhapatnam, India
Email: uma.nitw@gmail.com

Dharmala Venkata Padmaja
Department of Mechanical Engineering
BABA Institute of Technology and Sciences, Visakhapatnam, India
Email: padmajady@gmail.com

Abstract- Global warming has become a worldwide concern, which is highly contributed by the emissions of IC engine combustion products. Inflation in fuel prices as well and shortage of its supply due to unsustainable energy situations world-wide owing to unequal distribution of natural wealth as well as geographical, environmental and economic status are also another concern to look for an alternative source for this. Ever increasing drift of energy consumption is due to over population, transportation and luxurious life style has motivated researches to carry out research on renewable sources as sustainable alternative. In this study renewable, carbon neutral, non-toxic, essentially free of sulphur, bio-lubricants are synthesized as alternative for fossil fuels. Trimethylol propane esters, neopentyl glycol esters, pentaerythritol esters have been developed from Neem oil as lubricant base stocks. The developed base stocks were evaluated for their physico-chemical and lubricant properties. Trimethylol propane esters were found to be better lubricants compared to other esters synthesized.

Keywords – Calophyllum oil, esters, bio-lubricant.

1. INTRODUCTION

In present days utilization of crude oil was increased due to increased needs with growing technology and population. Crude oil is non-renewable and biodegradable. Lubricant property in vegetable oils is enhanced by converting them to polyol TMP ester [1]. The hydraulic fluids based TMP esters of rapeseed oil had good cold stability, friction and wear characteristics, and resistance against oxidation at elevated temperatures [2]. The production of polyol esters by transesterification of Jatropha oil with methanol and trimethylol propane. The effects of temperature, pressure, ratio of Jatropha oil methyl ester: trimethylol propane and catalyst amount were studied in optimization experiment. [3]. Preparation of polyol esters based on rapeseed oil, olive oil and lard with neopentyl glycol (NPG) and trimethylol propane (TMP). These polyol esters were characterised with physico chemical properties [4]. Merja Lämsä, et al., studied on 2-Ethyl-1-hexanol fatty acid esters from rapeseed oil by enzymatic transesterification [5]. The performance and exhaust emission characteristics of Neem methyl ester with diesel fuel [6]. Vegetable oil is biodegradable and shows comparable performance as a replacement for mineral oil lubricants [7]. Vegetable oil-based polyol esters have been produced in various ways from FA [8,9] and FAME [10]. However, in contrast, Waara et al. [11] found that pure synthetic ester (TMP-Oleate and TMP-C8- C10) in a sliding contact resulted in high wear rates and a lot of abrasive marks on the surface. Zulkifli NWM et al [12] has studied wear prevention characteristics of a palm oil-based TMP ester as an engine lubricant [13]. The Kinetics of Transesterification of Palm-based methyl esters with Trimethylol propane. Polyol esters synthesized are eco-friendly and biodegradable which are fire resistance and have good lubricating properties [14]. Karuna et al., has studied the polyol esters of rubber seed oil as a potential source for hydraulic oils and thumba seed oil as potential bio lubricant base stocks. Polyol esters of 10-undecenoic acid and undecanoic acid are potential lubricant base stock. [15]. Tamanu based biodiesel used as non conversational fuels in compression ignition engine without any need for modification in engine design [16]. Tribiological characteristics of Neem based TMP ester as a better

Analysis of Wind Turbine Blade of Composite Material Subjected to Different Loads by Identifying its Natural Frequency

Dharmala Venkata Padmaja

Department of Mechanical Engineering,

BABA Institute of Technology and Sciences, Visakhapatnam, India
padmajadv@gmail.com

Ch Uma Chaitanya

Department of Mechanical Engineering,

BABA Institute of Technology and Sciences, Visakhapatnam, India
umanitw@gmail.com

Sireesha Andiboyinaa

Department of Mechanical Engineering,

BABA Institute of Technology and Sciences, Visakhapatnam, India.
sireesha.andiboyinaa04@gmail.com

Abstract: Blades are being developed in composite materials in order to achieve low-weight and high strength construction. Composite blades are made with unidirectional fibers parallel to the blade's long axis. This project examines the modal analysis techniques applied in experiments using a uniform and a stepped beam. These simplified shapes are representative of a wind turbine blade. Natural frequencies have been identified, therefore designers can ensure those natural frequencies will not be close to the frequency of the main excitation forces in order to avoid resonance.

The turbine blade is approximated by a cantilever. Therefore, it is fully constrained where attached to a turbine shaft/hub. The specimens i.e, uniform beam is modeled in CATIA and analysis is done in ANSYS. Analysis of static, modal, harmonic and transient are carried out. The results obtained from analysis of uniform beam by changing material with steel and carbon epoxy is compared.

Keywords: Wind Turbine Blade, Natural Frequency, Composite Materials, Failure Analysis

1. INTRODUCTION

Recently, wind energy has become one of the most important forms of alternate energy sources. In a wind power production system, turbine blades are used to convert wind energy to electric energy. In general, turbine blades are required to be light in weight with high strength to withstand extreme wind loads and have a little mass moment of inertia so that they can start to rotate at low wind speed. On the other hand, a wind power system has to survive severe environment for at least 20 years. During its lifetime, a wind blade may experience strong wind with speed as high as 60m/s and in such condition, the blade should have enough strength to sustain the extreme wind load. By considering these requirements, the reliability of wind turbine blade has become an important research work. Recently, many research papers have been devoted to the study of different aspects of wind turbine blades including design, structural and theoretical analysis at different wind loads. A method established on the basis of ANSYS software on two shapes of blade and two different composite materials. Analysis is done on static, modal, harmonic and transient.

2. WIND BLADE CHARACTERISTICS

The shape of the wind turbine blade is considered as uniform beam. The wind turbine blade is modeled in CATIA and then it is imported to ANSYS with

Principal
Baba Institute of Technology and Sciences
Bakkannapalem (V), Madhurawada (P)
Visakhapatnam

Analysis of Wind Turbine Blade of Composite Material Subjected to Different Loads by Identifying its Natural Frequency

Dharmala Venkata Padmaja
Department of Mechanical Engineering,
BABA Institute of Technology and Sciences, Visakhapatnam, India
padmajadv@gmail.com

Ch Uma Chaitanya
Department of Mechanical Engineering,
BABA Institute of Technology and Sciences, Visakhapatnam, India
umanitw@gmail.com

Sireesha Andiboyinaa
Department of Mechanical Engineering,
BABA Institute of Technology and Sciences, Visakhapatnam, India.
sireesha.andiboyinaa04@gmail.com

Abstract: Blades are being developed in composite materials in order to achieve low-weight and high strength construction. Composite blades are made with unidirectional fibers parallel to the blade's long axis. This project examines the modal analysis techniques applied in experiments using a uniform and a stepped beam. These simplified shapes are representative of a wind turbine blade. Natural frequencies have been identified, therefore designers can ensure those natural frequencies will not be close to the frequency of the main excitation forces in order to avoid resonance.

The turbine blade is approximated by a cantilever. Therefore, it is fully constrained where attached to a turbine shaft/hub. The specimens i.e., uniform beam is modeled in CATIA and analysis is done in ANSYS. Analysis of static, modal, harmonic and transient are carried out. The results obtained from analysis of uniform beam by changing material with steel and carbon epoxy is compared.

Keywords: Wind Turbine Blade, Natural Frequency, Composite Materials, Failure Analysis

1. INTRODUCTION

Recently, wind energy has become one of the most important forms of alternate energy sources. In a wind power production system, turbine blades are used to convert wind energy to electric energy. In general, turbine blades are required to be light in weight with high strength to withstand extreme wind loads and have a little mass moment of inertia so that they can start to rotate at low wind speed. On the other hand, a wind power system has to survive severe environment for at least 20 years. During its lifetime, a wind blade may experience strong wind with speed as high as 60m/s and in such condition, the blade should have enough strength to sustain the extreme wind load. By considering these requirements, the reliability of wind turbine blade has become an important research work. Recently, many research papers have been devoted to the study of different aspects of wind turbine blades including design, structural and theoretical analysis at different wind loads. A method established on the basis of ANSYS software on two shapes of blade and two different composite materials. Analysis is done on static, modal, harmonic and transient.

2. WIND BLADE CHARACTERISTICS

The shape of the wind turbine blade is considered as uniform beam. The wind turbine blade is modeled in CATIA and then it is imported to ANSYS software with **PRINCIPAL** BABA Institute of Technology and Sciences Bakkannapalem (V), Madhurawada (P) Visakhapatnam

Analysis of Wind Turbine Blade of Composite Material Subjected to Different Loads by Identifying its Natural Frequency

Dharmala Venkata Padmaja
Department of Mechanical Engineering,
BABA Institute of Technology and Sciences, Visakhapatnam, India
padmajadv@gmail.com

Ch Uma Chaitanya
Department of Mechanical Engineering,
BABA Institute of Technology and Sciences, Visakhapatnam, India
umanitw@gmail.com

Sireesha Andiboyinaa
Department of Mechanical Engineering,
BABA Institute of Technology and Sciences, Visakhapatnam, India.
sireesha.andiboyinaa04@gmail.com

Abstract: Blades are being developed in composite materials in order to achieve low-weight and high strength construction. Composite blades are made with unidirectional fibers parallel to the blade's long axis. This project examines the modal analysis techniques applied in experiments using a uniform and a stepped beam. These simplified shapes are representative of a wind turbine blade. Natural frequencies have been identified, therefore designers can ensure those natural frequencies will not be close to the frequency of the main excitation forces in order to avoid resonance.

The turbine blade is approximated by a cantilever. Therefore, it is fully constrained where attached to a turbine shaft/hub. The specimens i.e., uniform beam is modeled in CATIA and analysis is done in ANSYS. Analysis of static, modal, harmonic and transient are carried out. The results obtained from analysis of uniform beam by changing material with steel and carbon epoxy is compared.

Keywords: Wind Turbine Blade, Natural Frequency, Composite Materials, Failure Analysis

1. INTRODUCTION

Recently, wind energy has become one of the most important forms of alternate energy sources. In a wind power production system, turbine blades are used to convert wind energy to electric energy. In general, turbine blades are required to be light in weight with high strength to withstand extreme wind loads and have a little mass moment of inertia so that they can start to rotate at low wind speed. On the other hand, a wind power system has to survive severe environment for at least 20 years. During its lifetime, a wind blade may experience strong wind with speed as high as 60m/s and in such condition, the blade should have enough strength to sustain the extreme wind load. By considering these requirements, the reliability of wind turbine blade has become an important research work. Recently, many research papers have been devoted to the study of different aspects of wind turbine blades including design, structural and theoretical analysis at different wind loads. A method established on the basis of ANSYS software on two shapes of blade and two different composite materials. Analysis is done on static, modal, harmonic and transient.

2. WIND BLADE CHARACTERISTICS

The shape of the wind turbine blade is considered as uniform beam. The wind turbine blade is modeled in CATIA and then it is imported to ANSYS Software with

PRINCIPAL
BABA Institute of Technology and Sciences
Bakkannapalem (V), Madhurawada (P)
Visakhapatnam

Conversion of Waste Plastic into Burning Oil: Review

R.Rajesh

*Assistant Professor, Department of Mechanical Engineering
Baba Institute of Technology and sciences, Visakhapatnam*

S.M.Idris

*Assistant Professor, Department of Mechanical Engineering
Baba Institute of Technology and sciences, Visakhapatnam
syedmahaboobidris@gmail.com*

Abstract: Now a day's utilization of plastics are day by day increasing. These plastics are made from crude oils through polymerization process or poly condensation process. After utilization of plastic materials those are treated as waste plastic once it enters in to the earth can't be degradable easily these are causes for very dangerous problems to human beings, all living animals in the earth. There are number of techniques to reduce this plastic pollution. In that first technique is to dump waste plastic in to the sea when this happens the living things which are in the sea will swallow so that those life will be decrease. another technique is to dump waste plastic into lands due to this the fertility of earth will decreases another technique is to burn this waste plastic with in the atmosphere this is very dangerous. because once plastic burnt this will produce very harmful gases these are causes for health problems for humans.

To avoid these problems one technique is generally used for to convert this waste plastic in to burning fuel that technique is pyrolysis process. In this process waste plastic is generally heated in the absence of oxygen and gets burning oil. The main objective of this paper is to explain about pyrolysis technique.

Keywords: waste plastic, pyrolysis process, burning oil.

1. INTRODUCTION

Plastics play an vital part in day- nowadays life. It is interesting fabric since of their durability, light weight, resistance to water and chemicals, safe to warm and cold, moo electrical and warm conductivity, ease of creation, exceptional color extend, more plan adaptability, strength and vitality effectiveness. Due to over properties it is utilized in bundling materials, agribusiness, development, separator, car segment, electronic gadgets, materials and sports hardware and toys. Plastics constitutes in two primary categories. It is thermoplastics and thermoset plastics. Thermoplastics make up 80% of the plastics and thermoset plastics make up of remaining 20% of plastics delivered nowadays (Birley et al, 1988), etc. Thermo plastics can re-melt or remold and so it recyclable effectively but thermoset plastics cannot re-melt or reshape and so it is troublesome to reusing. Utilize of diverse sort of a few thermo plastics is given in table underneath. Plastics are generally cheap, simple accessible, simple to fabricate and their flexibility supplant to customary materials. Plastic squander


PRINCIPAL
Baba Institute of Technology and Sciences
Bakkannapalem (V), Madhurawada (P)
Visakhapatnam

Conversion of Waste Plastic into Burning Oil: Review

R.Rajesh

*Assistant Professor, Department of Mechanical Engineering
Baba Institute of Technology and sciences, Visakhapatnam*

S.M.Idris

*Assistant Professor, Department of Mechanical Engineering
Baba Institute of Technology and sciences, Visakhapatnam
syedmahaboobidris@gmail.com*

Abstract: Now a day's utilization of plastics are day by day increasing. These plastics are made from crude oils through polymerization process or poly condensation process. After utilization of plastic materials those are treated as waste plastic once it enters in to the earth can't be degradable easily these are causes for very dangerous problems to human beings, all living animals in the earth. There are number of techniques to reduce this plastic pollution. In that first technique is to dump waste plastic in to the sea when this happens the living things which are in the sea will swallow so that those life will be decrease. another technique is to dump waste plastic into lands due to this the fertility of earth will decreases another technique is to burn this waste plastic with in the atmosphere this is very dangerous. because once plastic burnt this will produce very harmful gases these are causes for health problems for humans.

To avoid these problems one technique is generally used for to convert this waste plastic in to burning fuel that technique is pyrolysis process. In this process waste plastic is generally heated in the absence of oxygen and gets burning oil. The main objective of this paper is to explain about pyrolysis technique.

Keywords: waste plastic, pyrolysis process, burning oil.

1. INTRODUCTION

Plastics play an vital part in day- nowadays life. It is interesting fabric since of their durability, light weight, resistance to water and chemicals, safe to warm and cold, moo electrical and warm conductivity, ease of creation, exceptional color extend, more plan adaptability, strength and vitality effectiveness. Dueto over propertiesitis utilized in bundling materials, agribusiness, development, separator, car segment, electronic gadgets, materials and sports hardware and toys. Plastics constitutes in two primary categories. It is thermoplastics and thermoset plastics. Thermoplastics make up 80% of the plastics and thermoset plastics make up of remaining 20% of plastics delivered nowadays (Birley et al, 1988), etc. Thermo plastics can re-melt or remold and so it recyclable effectively but thermoset plastics cannot re-melt or reshape and so it is troublesome to reusing. Utilize of diverse sort of a few thermo plastics is given in table1 underneath. Plastics are generally cheap, simple accessible, simple to fabricate andtheir flexibility supplant to customary materials. Plastic squander



PRINCIPAL
Baba Institute of Technology and Sciences
Bakkannapalem (V), Madhurawada (P)
Visakhapatnam